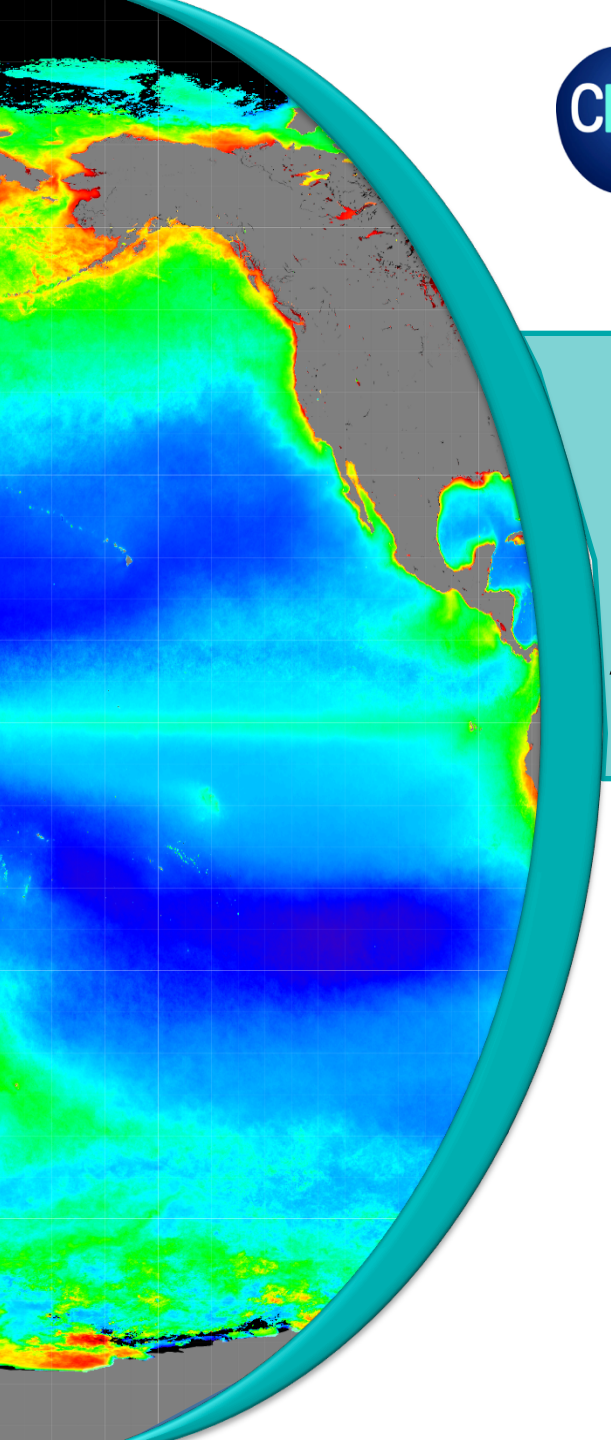


# Merging Ocean Color and physical Argo data to extend bio-optical properties to depth using neural networks:

## A 3D view of phytoplankton biomass and phytoplankton communities composition


**Raphaëlle Sauzède[1],** H. Claustre[1], J. Uitz[1], C. Jamet[2], G. Dall'Olmo[3], F. D'Ortenzio[1], B. Gentili[1], A. Poteau[1] and C. Schmechtig[1]

1: Laboratoire d'Océanographie de Villefranche, LOV
2: Laboratoire d'Océanologie et de Géosciences, LOG
3: Plymouth Marine Laboratory, PML



2016 OCEAN SCIENCES MEETING

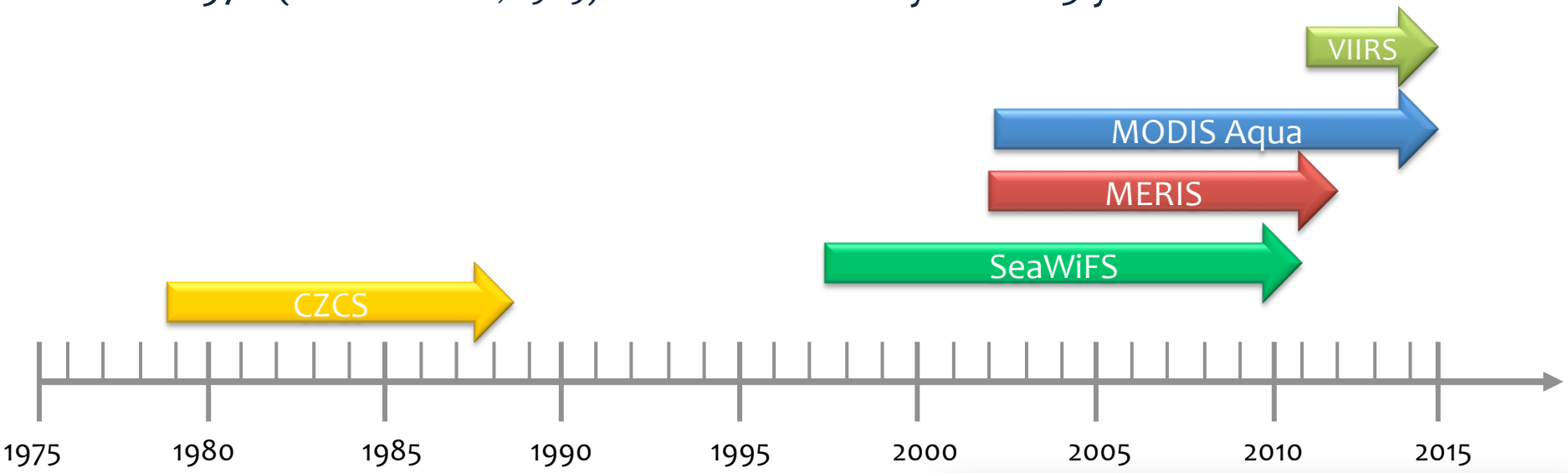
AGU • ASLO • The Oceanography Society

21–26 February • New Orleans, Louisiana, USA

Session ME52A

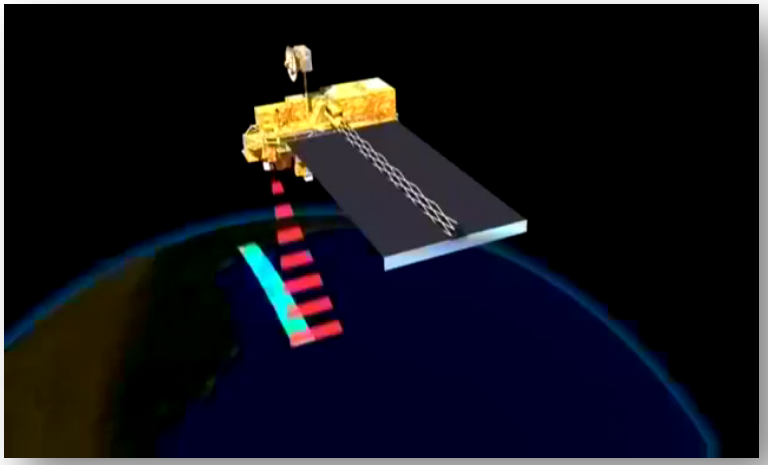
# Why extending surface bio-optical properties to depth

Several bio-optical proxies of the phytoplankton biomass are estimated from remote sensing of **ocean color** radiometry (McCluney, 1975) since the launch of CZCS sensor in 1978 (Feldman *et al.*, 1989) and continuously since 15 years




**High spatial and temporal resolution data**

→ Essential tools for the quantification of the phytoplankton biomass at global scale (McClain, 2009)



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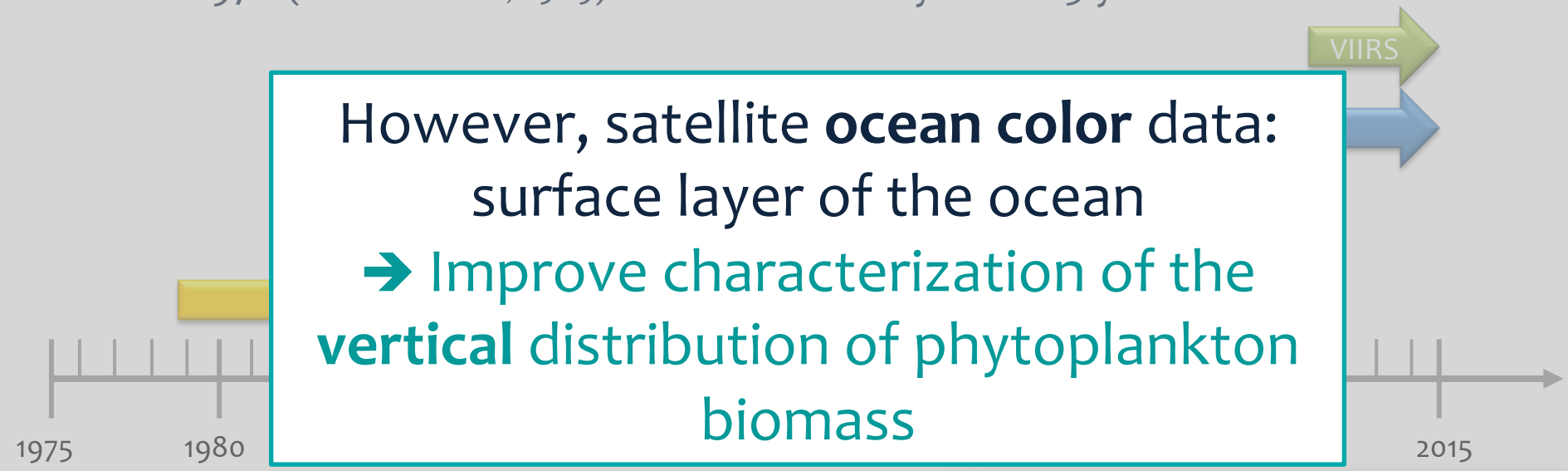
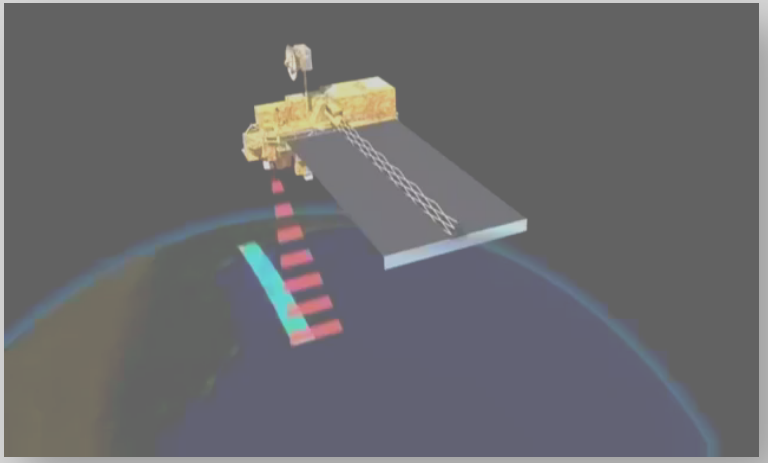



However, satellite **ocean color** data: surface layer of the ocean

➔ Improve characterization of the **vertical** distribution of phytoplankton biomass

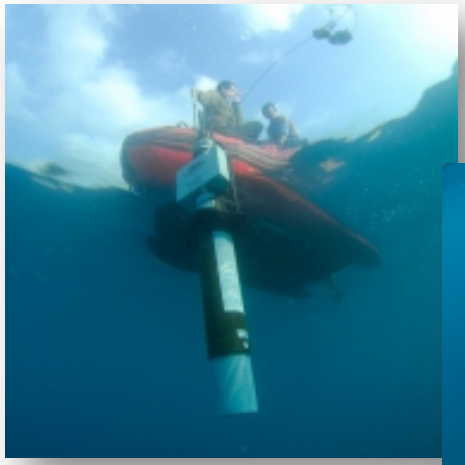
**High spatial and temporal resolution data**

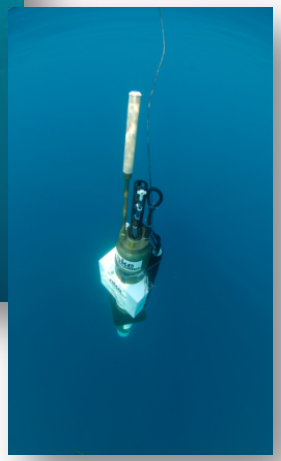
→ Essential tools for the quantification of the phytoplankton biomass at global scale (McClain, 2009)



# Global network to assess vertical distribution of oceanographic properties at high spatio-temporal resolution

## Profiling floats





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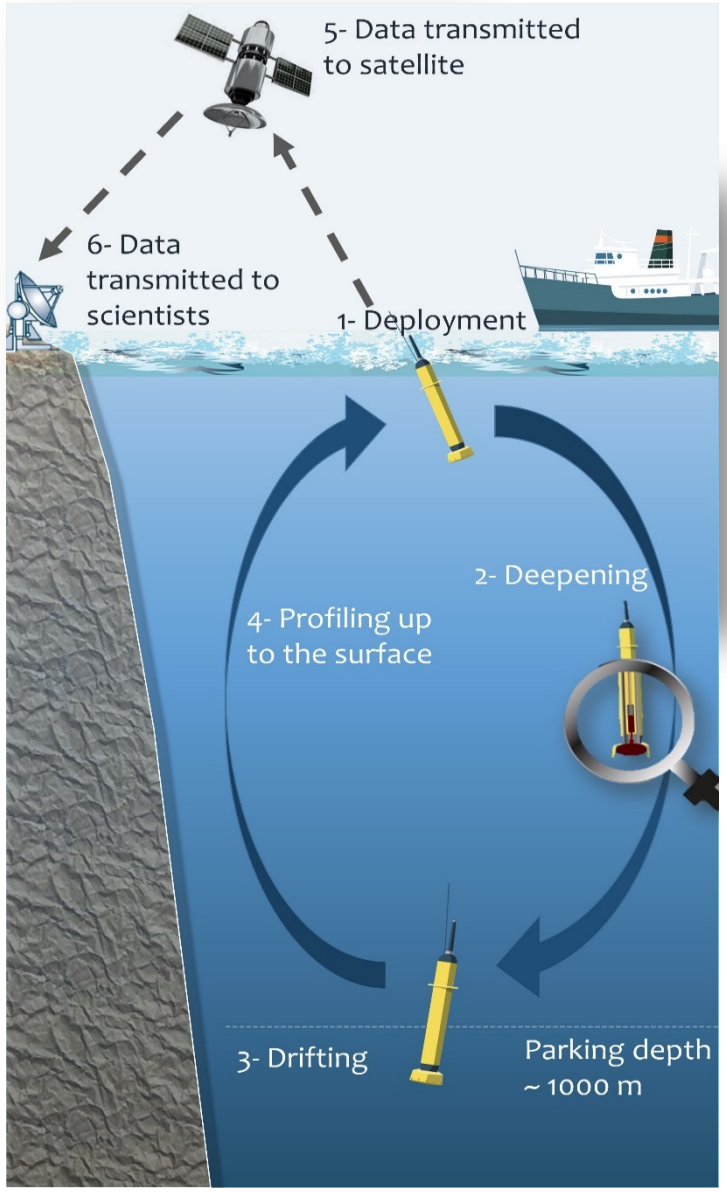


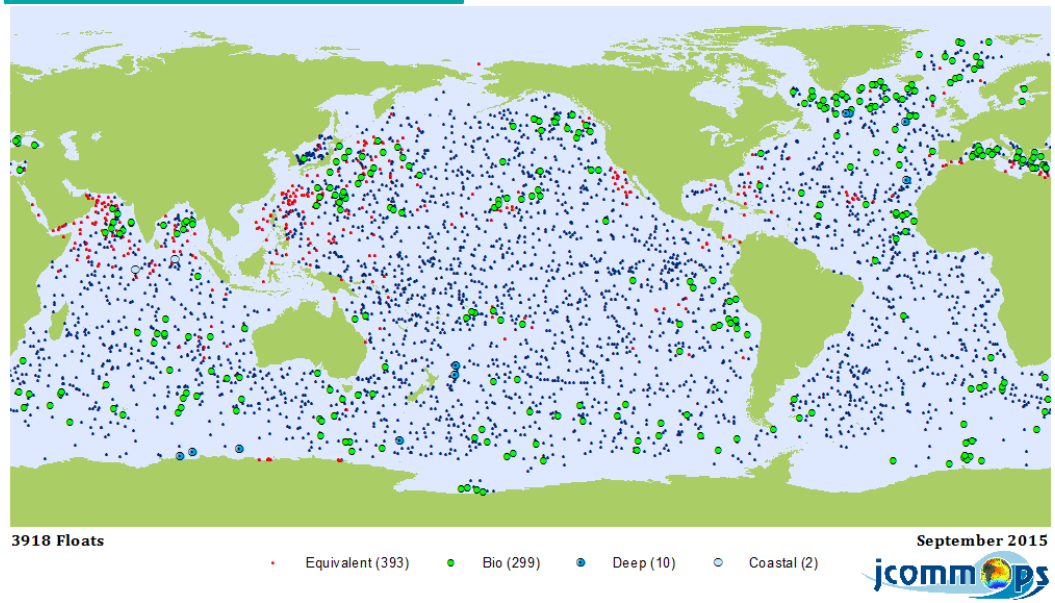

Measurements cycle of a profiling float

# Global network to assess vertical distribution of oceanographic properties at high spatio-temporal resolution

## Profiling floats

Temperature & salinity

Argo floats



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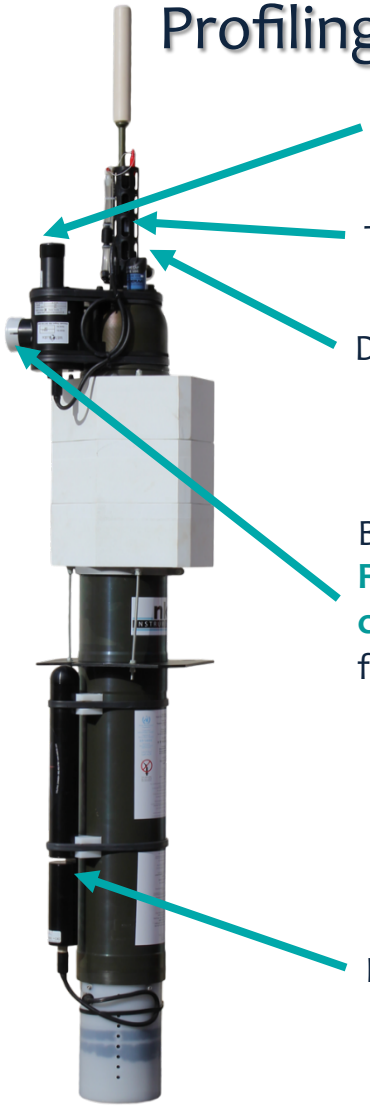


Radiometry: PAR, $E_d$

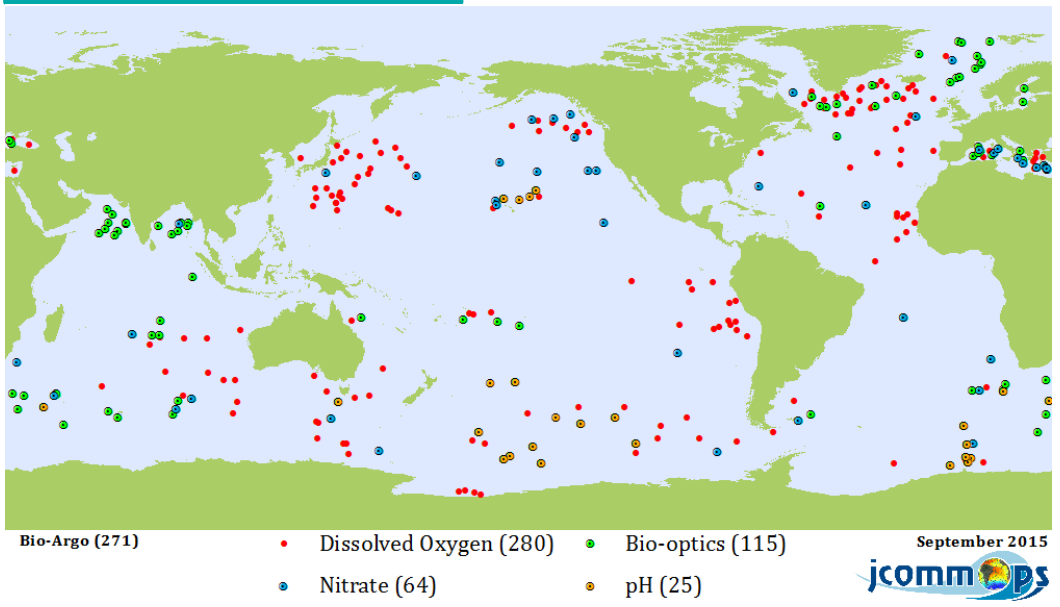
Temperature & salinity

Dissolved oxygen sensor

Bio-optical sensors: **Fluorescence of chlorophyll, $b_{bp}$** and fluorescence of CDOM

Nitrates sensor (SUNA)

## Bio-Argo floats



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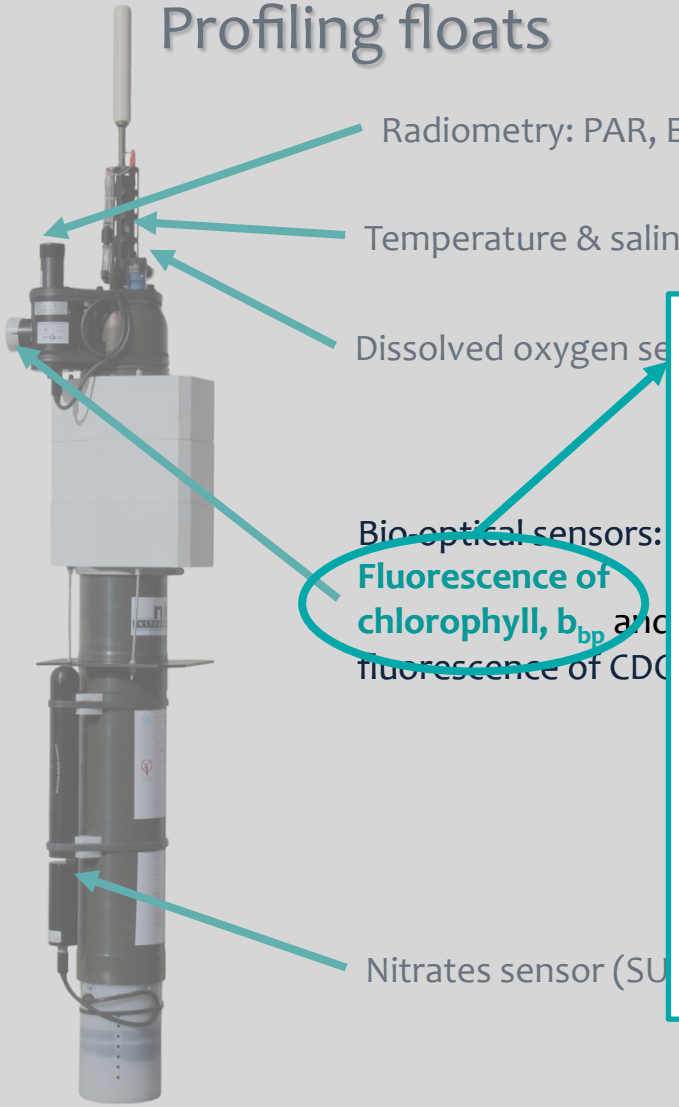
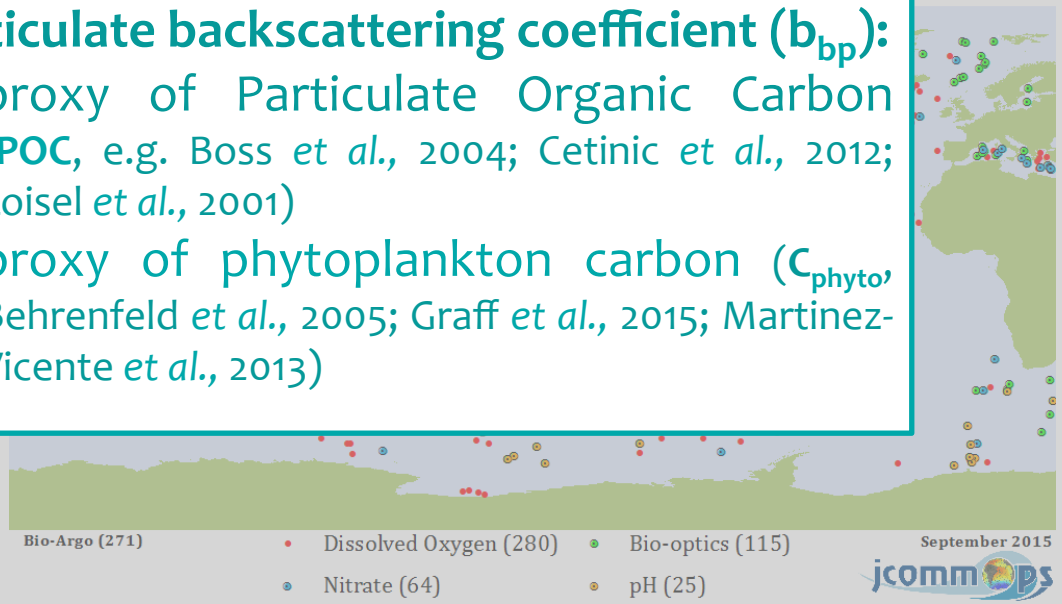



**Fluorescence of chlorophyll:**

- widely used proxy of chlorophyll *a* concentration (**Chl**, Lorenzen, 1966)

**Particulate backscattering coefficient ($b_{bp}$):**

- proxy of Particulate Organic Carbon (**POC**, e.g. Boss *et al.*, 2004; Cetinic *et al.*, 2012; Loisel *et al.*, 2001)
- proxy of phytoplankton carbon (**$C_{phyto}$**, Behrenfeld *et al.*, 2005; Graff *et al.*, 2015; Martinez-Vicente *et al.*, 2013)



# Bio-optical data :
# → good candidates for a 3D reconstruction of phytoplankton biomass at a global scale

Radiometry: PAR, $E_d$

Temperature & salinity

Dissolved oxygen se

Bio-optical sensors:
**Fluorescence of chlorophyll, $b_{bp}$** and
fluorescence of CDC

Nitrates sensor (SU

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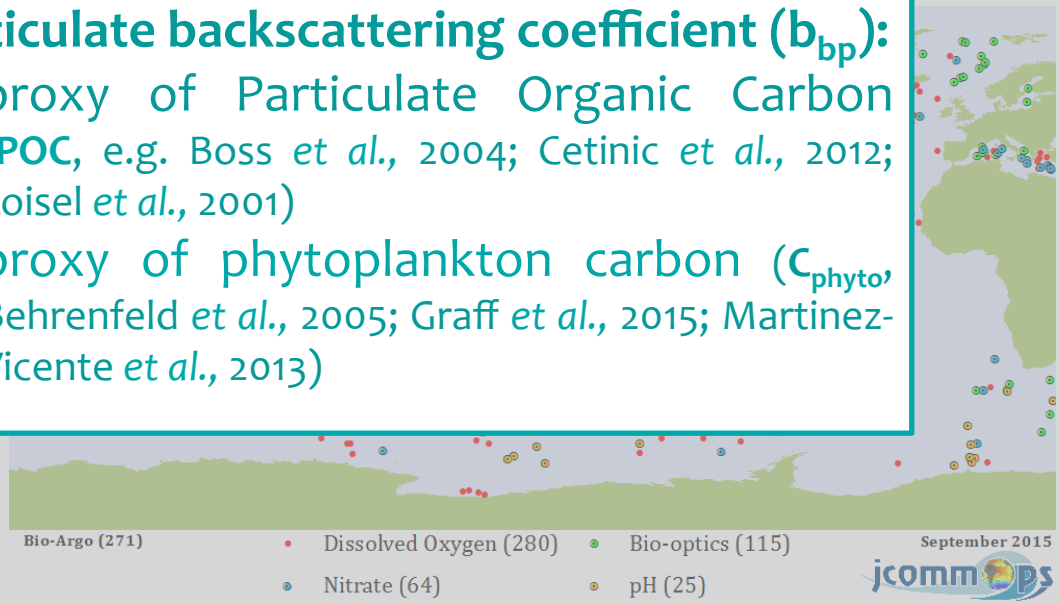
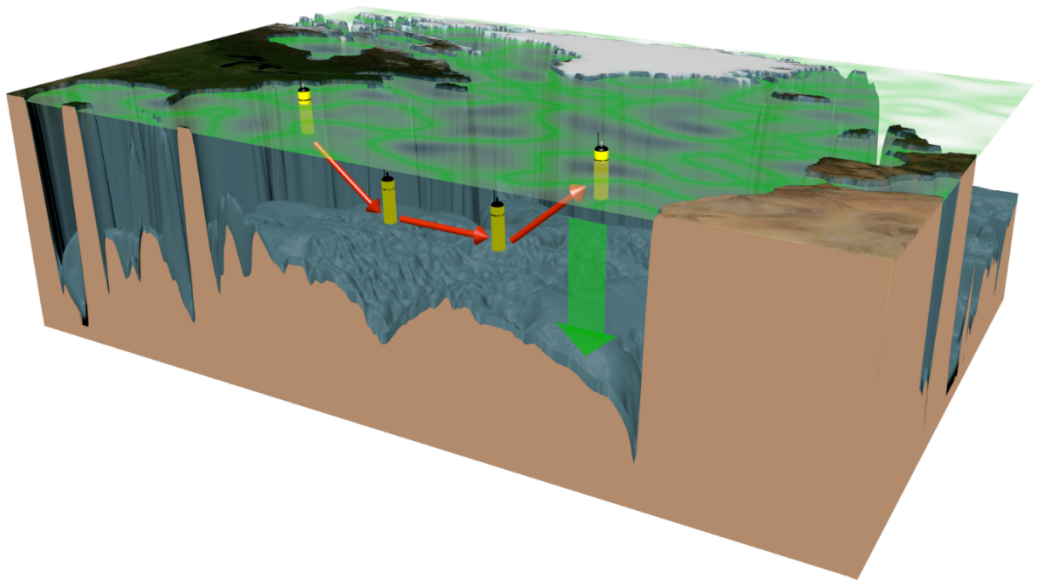


# MAIN OBJECTIVE

## How to extend surface bio-optical properties to depth from ocean color data?



**Physical state of the water column (from CTD profiles):**

- Influences nutrient and light availability for phytoplankton growth
- Available at high spatio-temporal frequency with Argo data

**Methodology: MLP for Multi-Layered Perceptron (neural network) :**

- Iterative statistical method of learning
- Universal approximators
- Complex, noisy and non-linear distributions

# OBJECTIVE

## To develop a neural network-based method using merged ocean color and Argo data to extend surface bio-optical properties to depth

Method after refer as **SOCA** (for Satellite Ocean Color and Argo data to vertical distribution of bio-optical properties)

**2 MLP-based parameterizations:**

### SOCA-BBP

for optical particulate backscattering coefficient ($b_{bp}$)

**Argo T/S profiles and ocean color matchup**

**→ vertical profiles of $b_{bp}$**

### SOCA-CHL

for chlorophyll-*a* concentration and phytoplankton size classes

**Argo T/S profiles and ocean color matchup**

**→ vertical profiles of Chl, micro-Chl, nano-Chl and pico-Chl**

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GLOBAL SCALE

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REGIONAL SCALE

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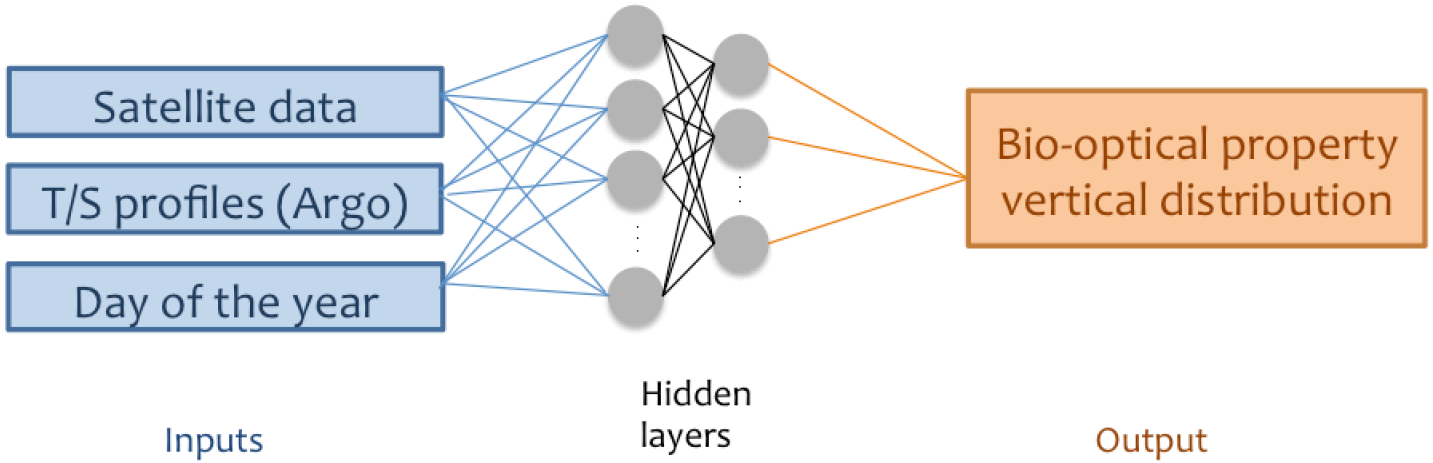
# SOCA-BBP architecture



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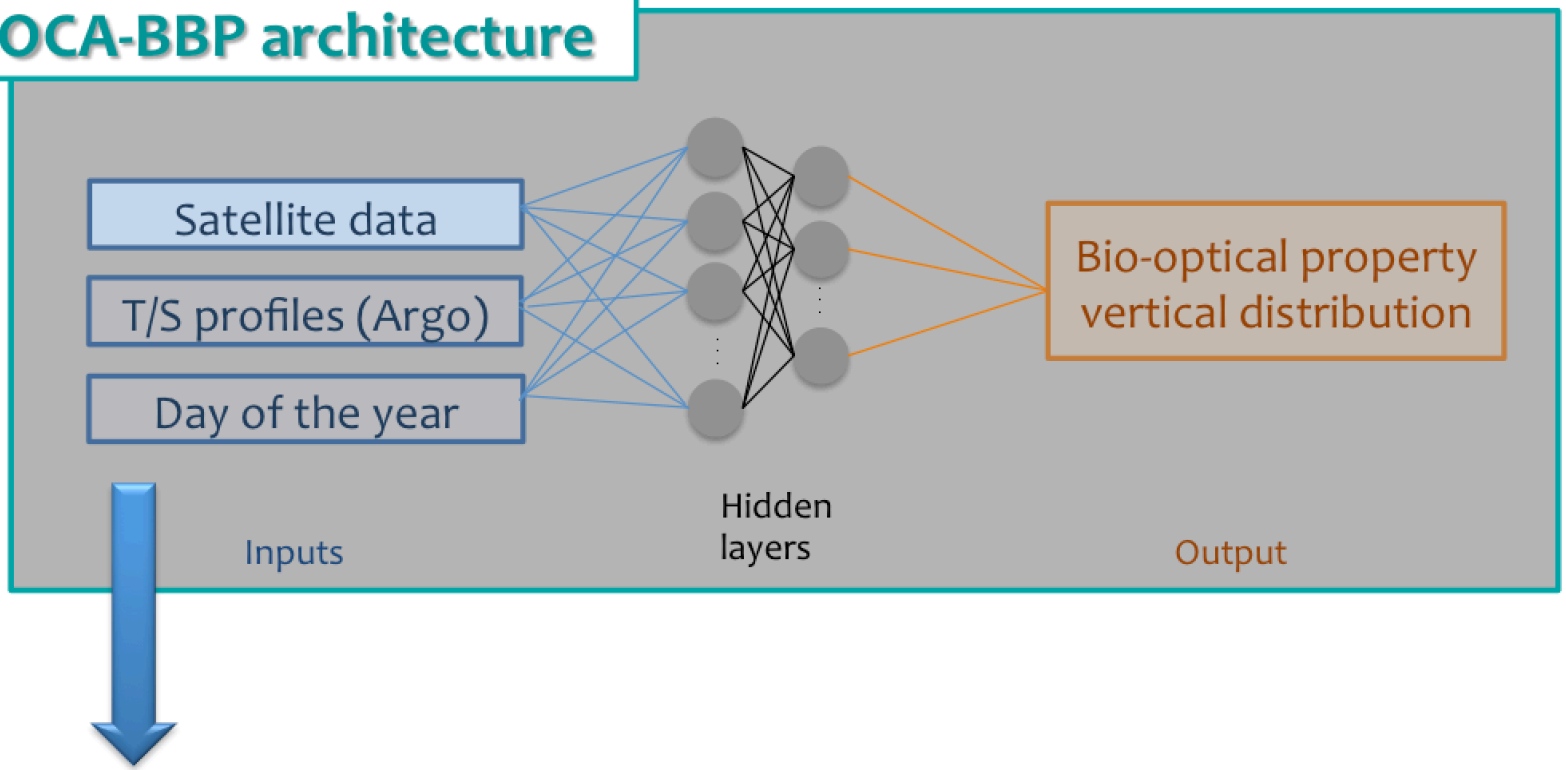
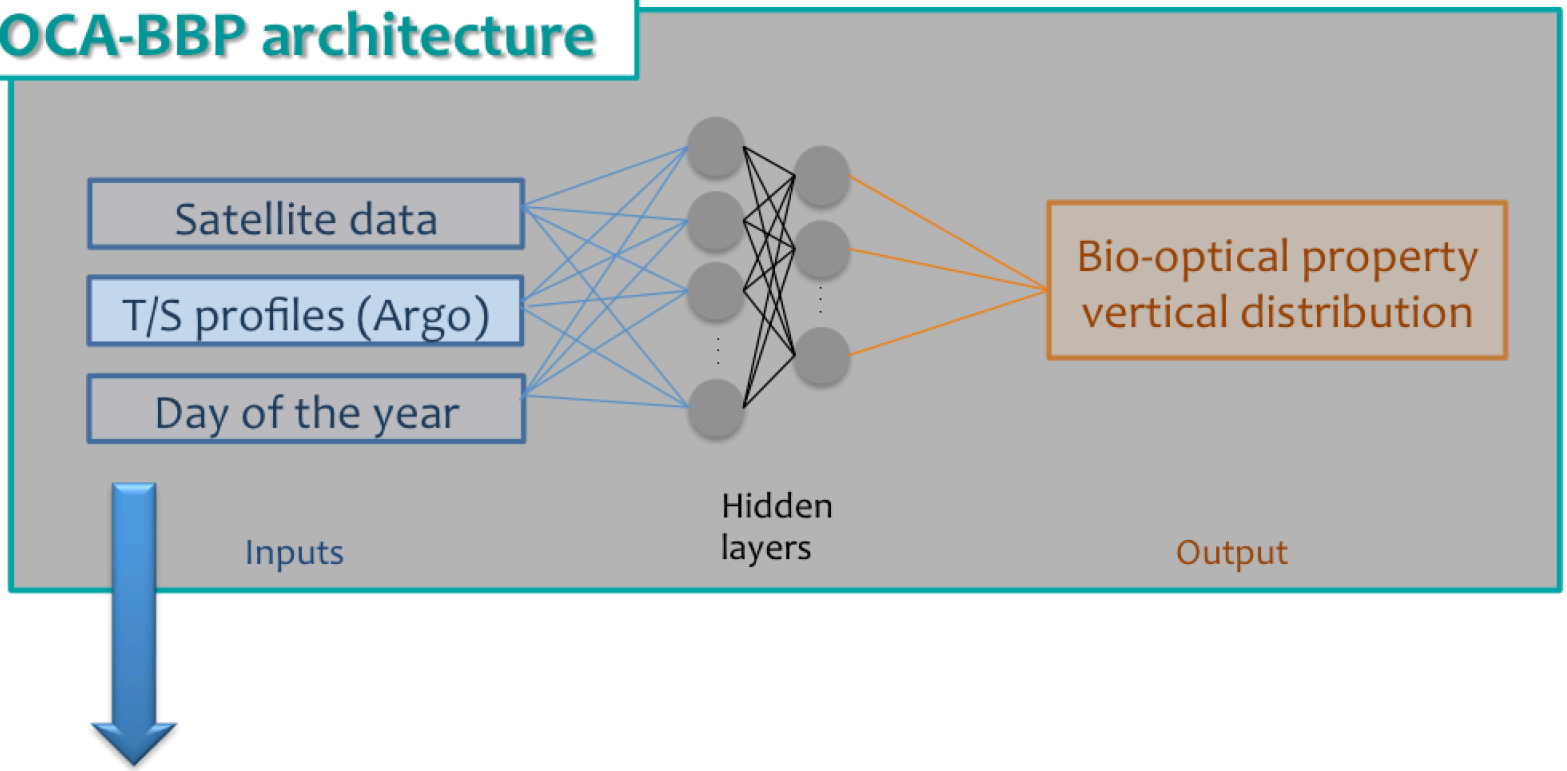



- log(**$b_{bp_sat}$(700)**) derived from the Quasi-Analytical Algorithm (QAA, Lee *et al.*, 2002) and log(**$Chl_{sat}$**)

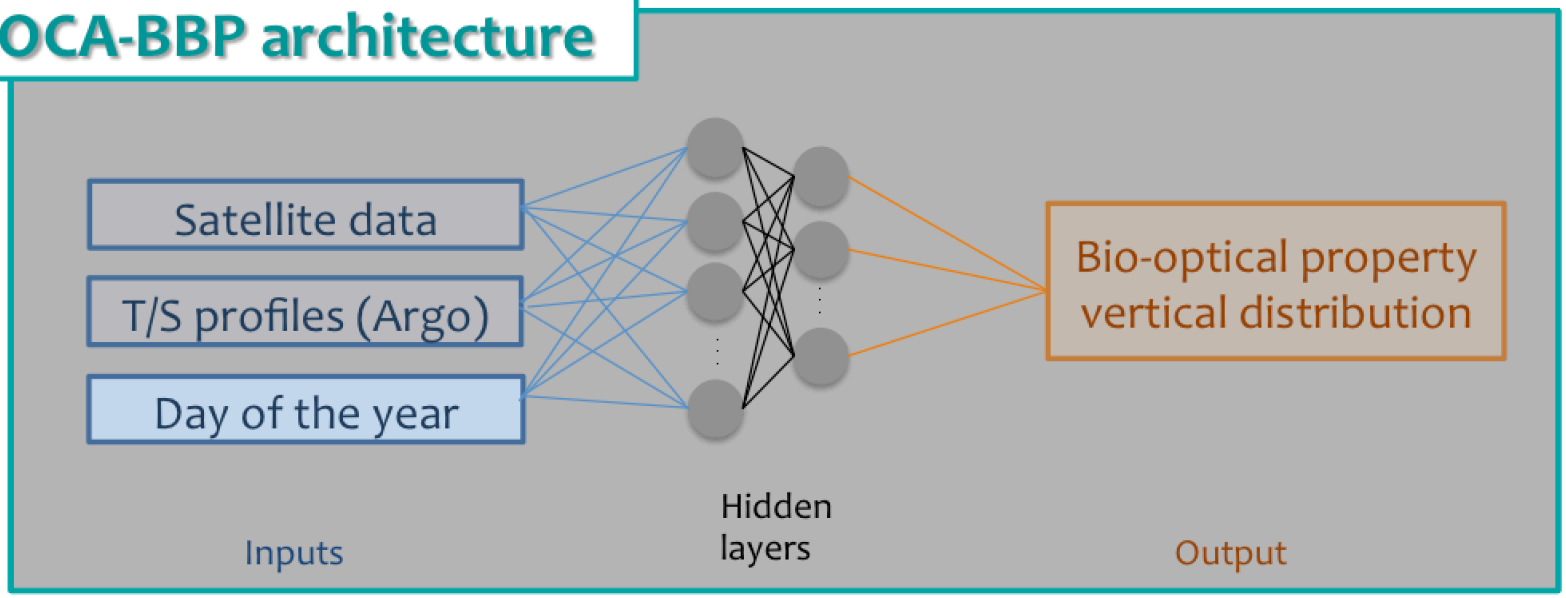
→ **From the satellite ocean color sensor MODIS Aqua**

# SOCA-BBP architecture




**Vertically-resolved physical properties derived from the Argo T/S profiles**

# SOCA-BBP architecture



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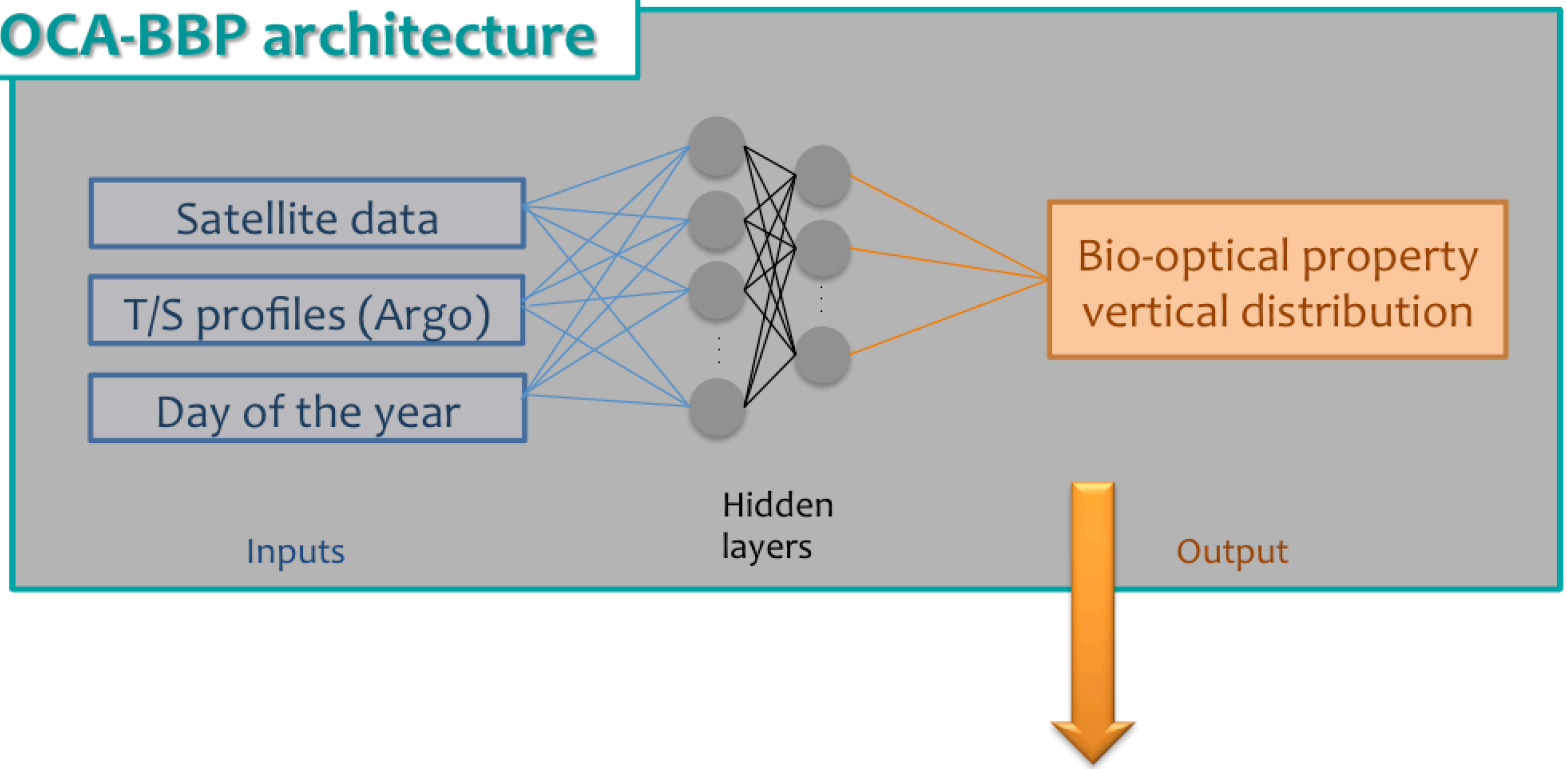
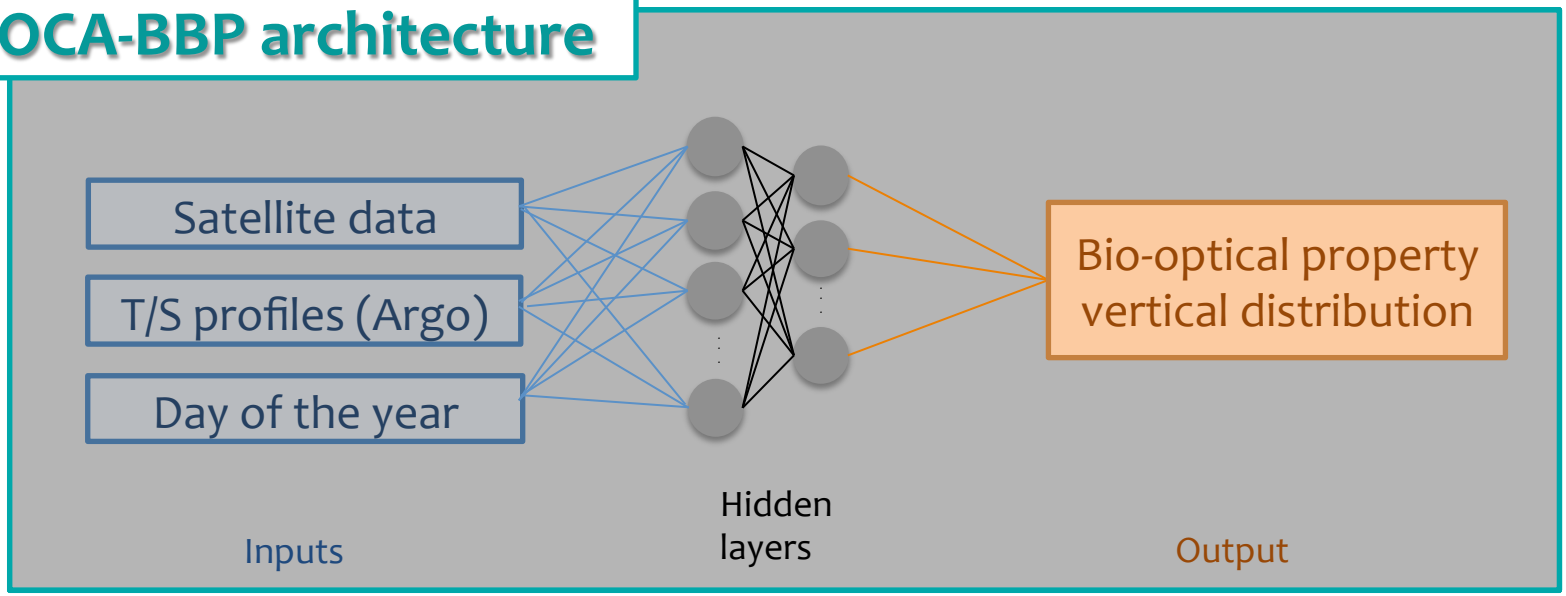



- 10 $b_{bp}(700)$ values for the water column simultaneously

# SOCA-BBP architecture



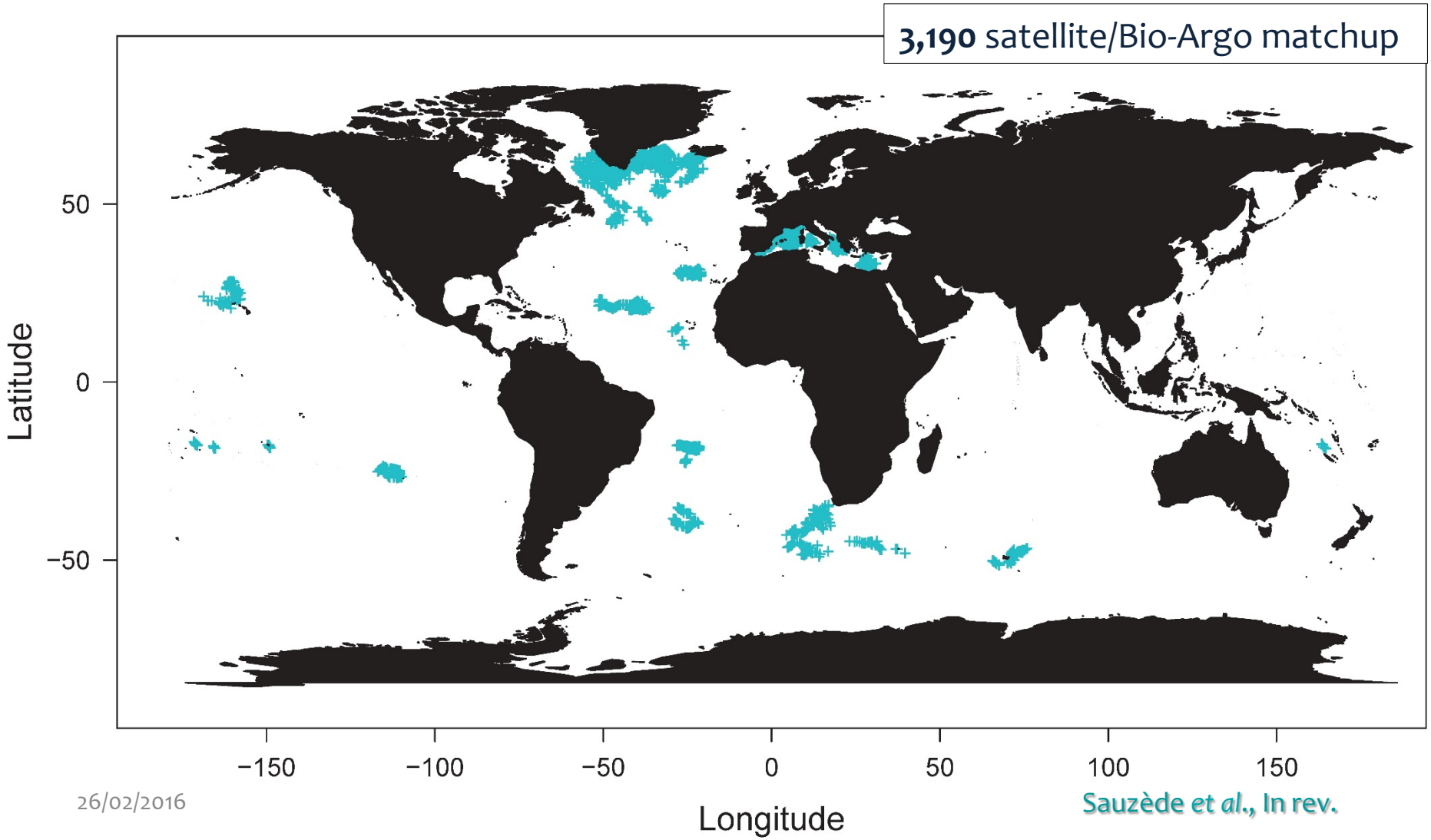

MLP trained and validated using essentially Bio-Argo floats data

Be careful!

**Argo** data (only T/S) used as inputs.
Bio-Argo floats data are only used as reference values to train and validate the MLP.

# MLP training and validation database

Concurrent profiles of density and $b_{bp}$ collected by **Bio-Argo floats,** concomitant with **satellite products** (i.e. matchups from 9-km/8-day MODIS Aqua)






Sauzède *et al.*, In rev.

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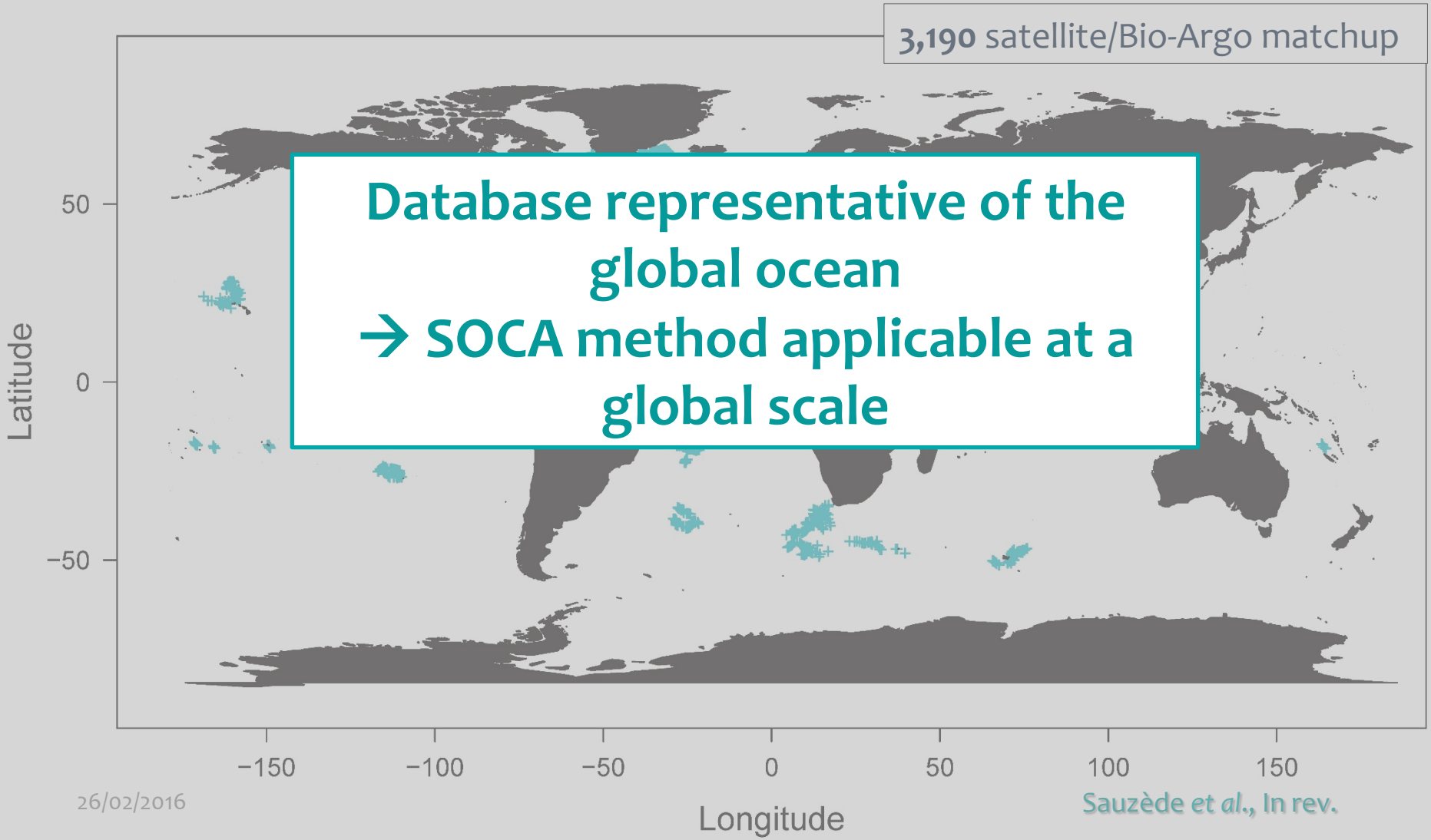


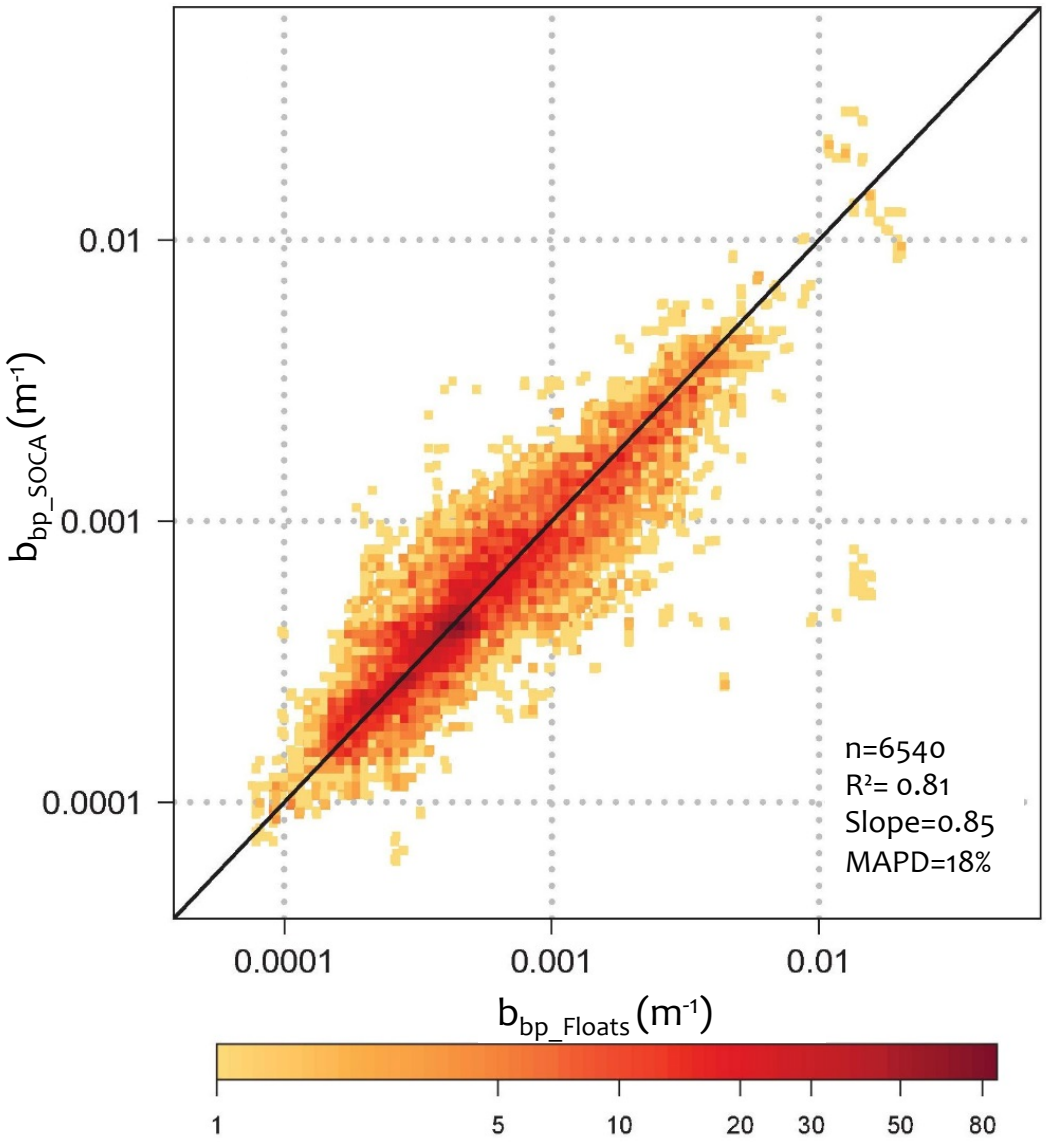



Sauzède *et al.*, In rev.

# Performance of $b_{bp}$ retrieval

## Validation:

→ 654 profiles from the 20% of the Bio-Argo database

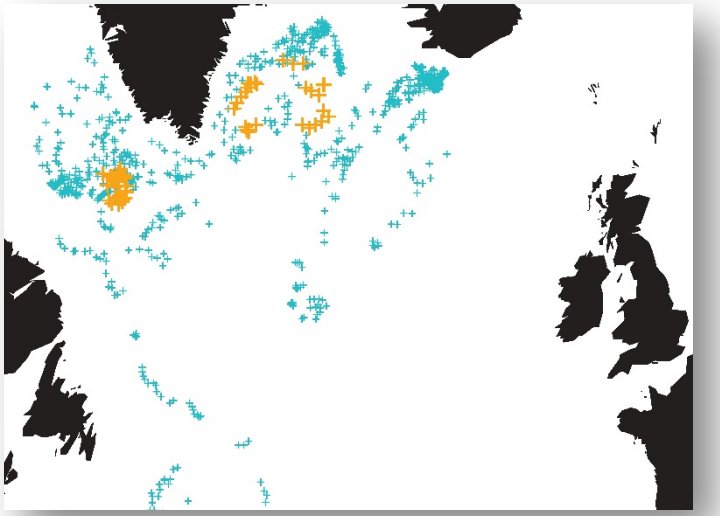



Sauzède *et al.*, In rev.

# Performance of $b_{bp}$ retrieval

validation using 2 time series from « independent » floats

North Atlantic Supbolar Gyre



Sauzède *et al.*, In rev.

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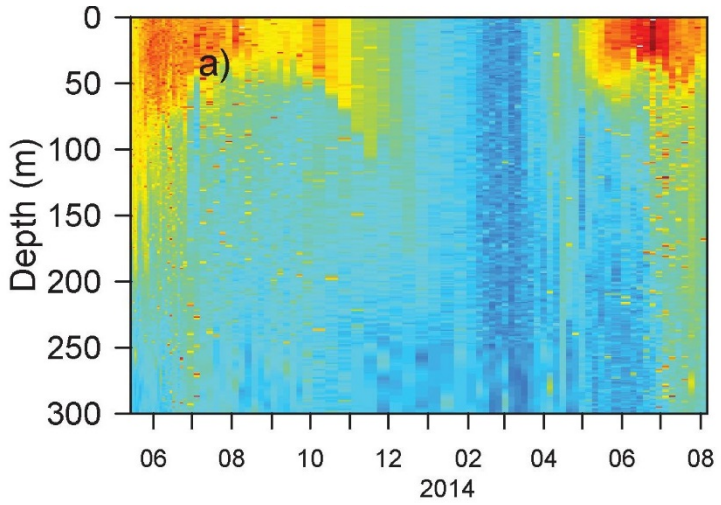
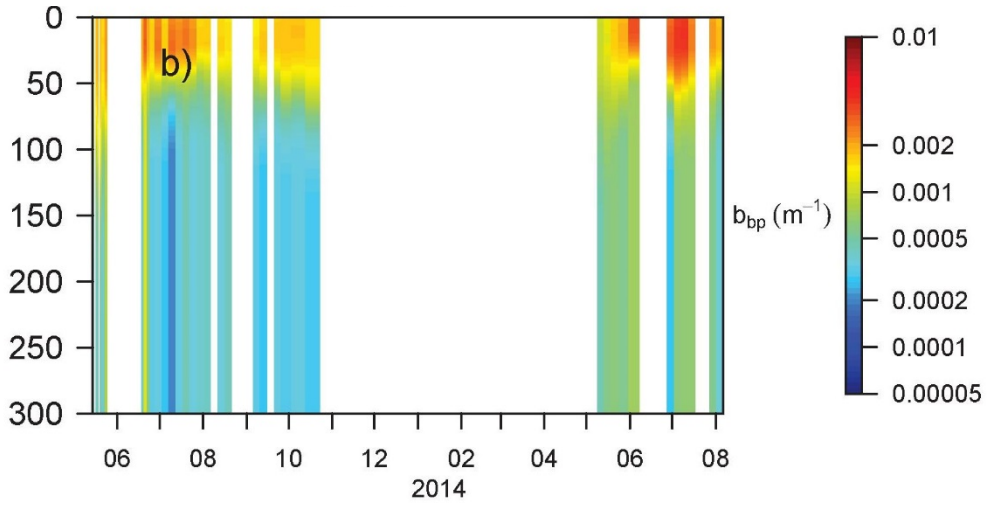
**$b_{bp}$ from Bio-Argo floats**




**$b_{bp}$ from SOCA:**

Satellite-Bio-Argo (only T/S) matchup

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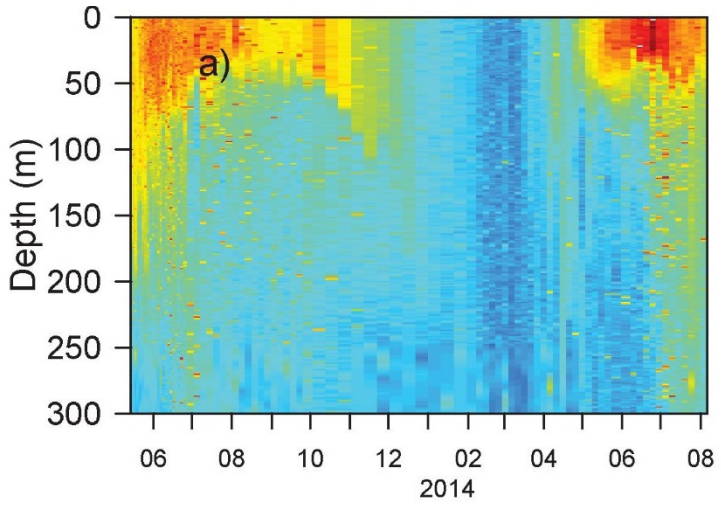


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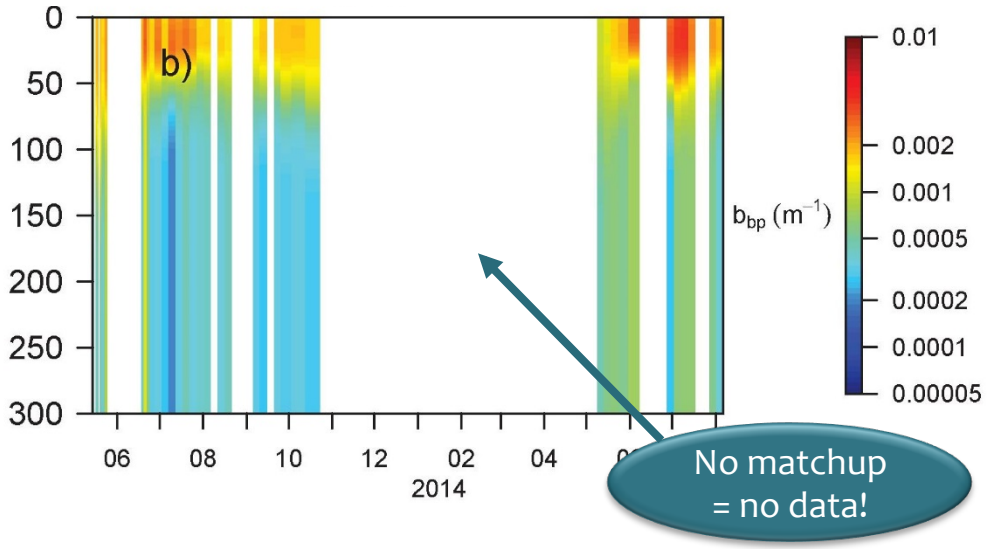


Sauzède *et al.*, In rev.

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North Atlantic Supbolar Gyre

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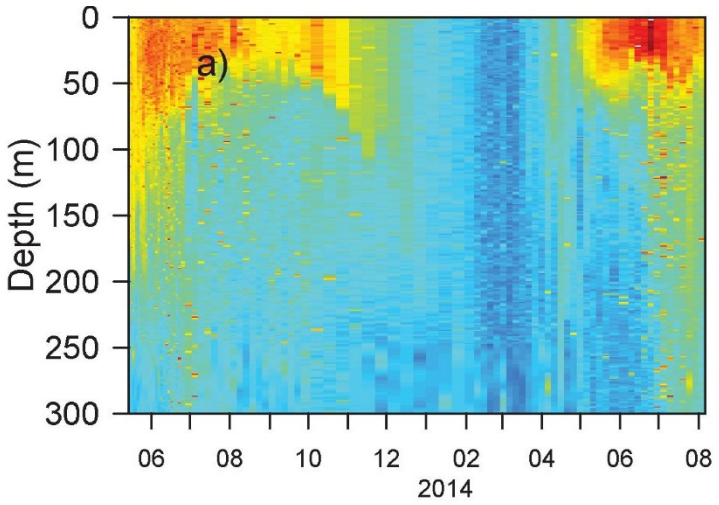


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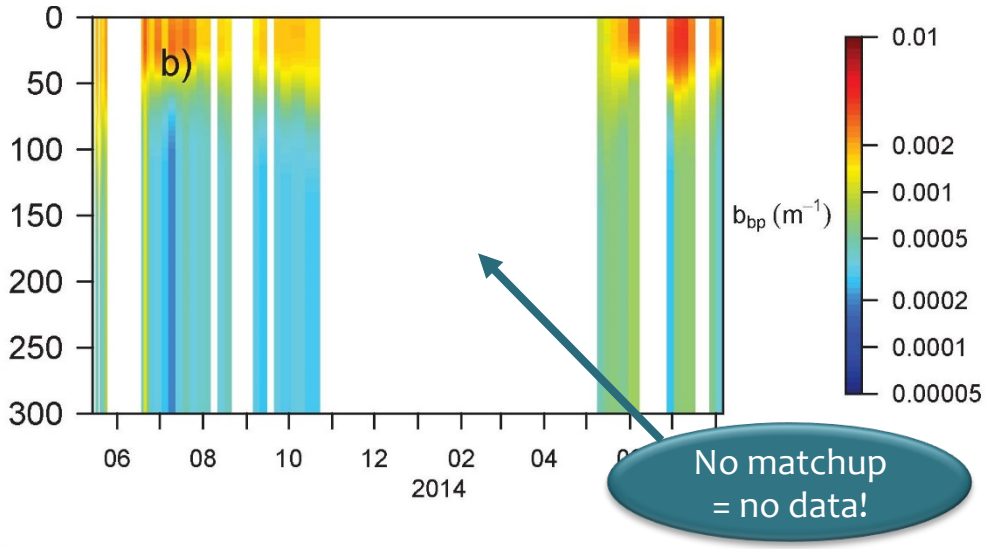



North Atlantic Subtropical Gyre

Sauzède *et al.*, In rev.

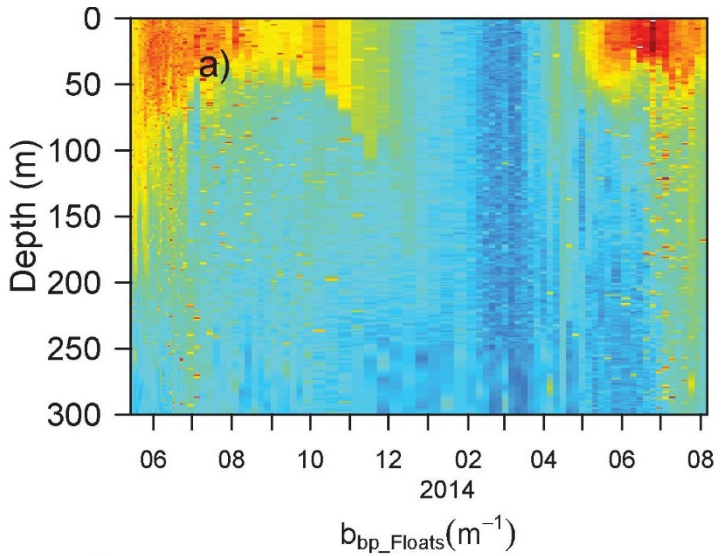
# Performance of $b_{bp}$ retrieval

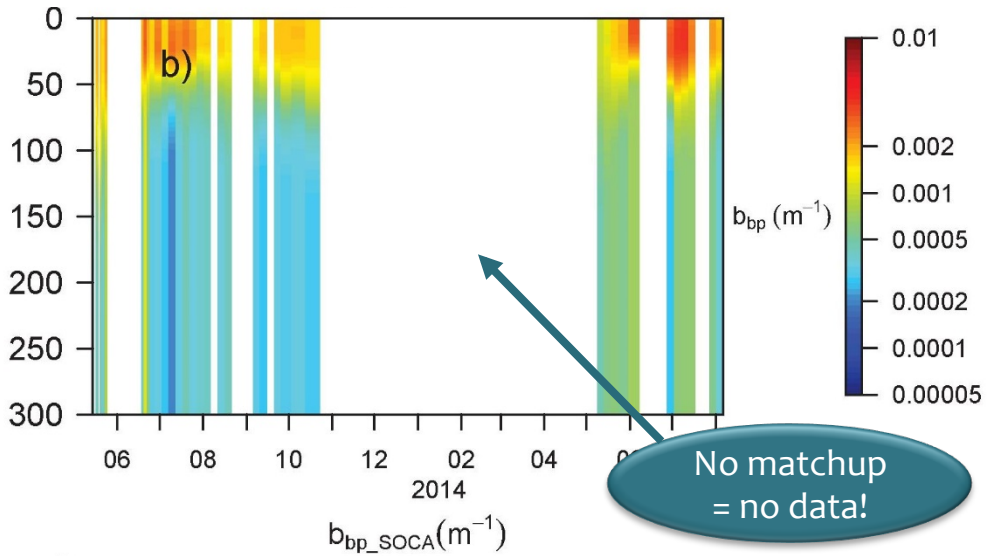
validation using 2 time series from « independent » floats

## $b_{bp}$ from Bio-Argo floats

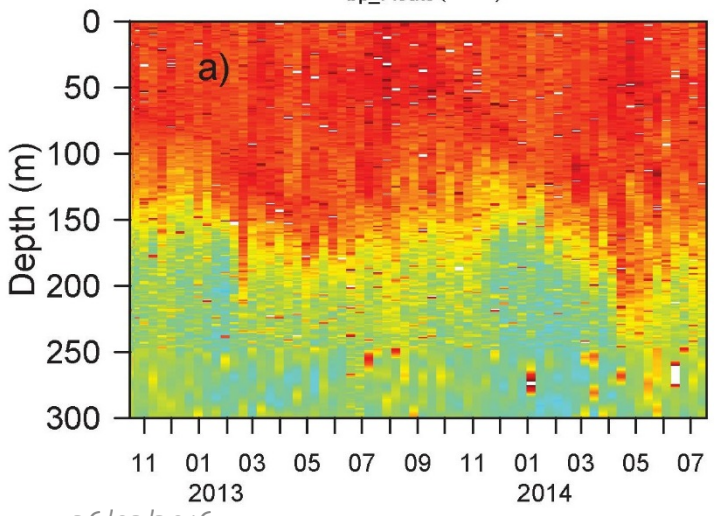
## $b_{bp}$ from SOCA:
Satellite-Bio-Argo (only T/S) matchup

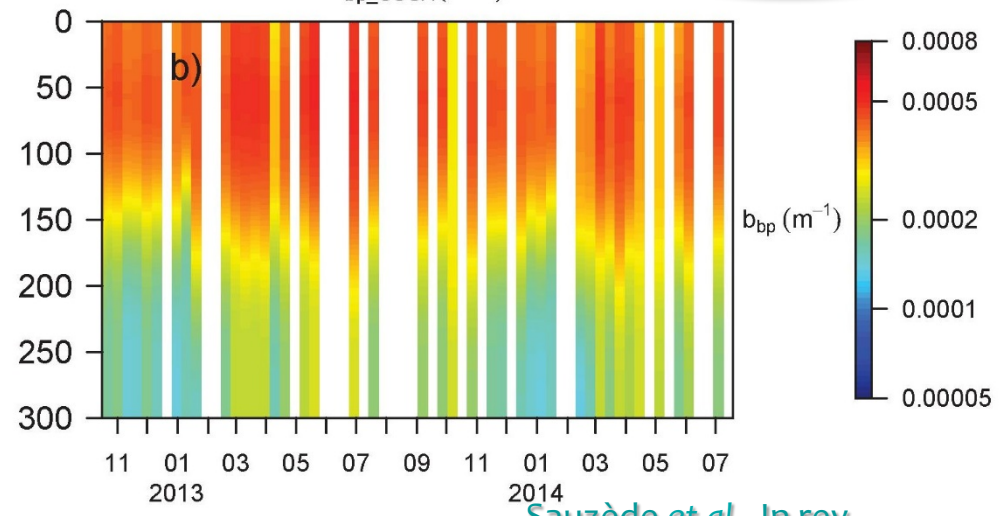
North Atlantic Supbolar Gyre




North Atlantic Subtropical Gyre



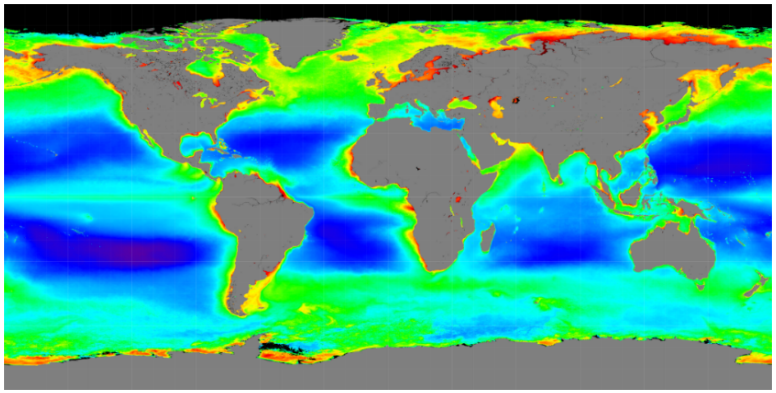




Sauzède *et al.*, In rev.

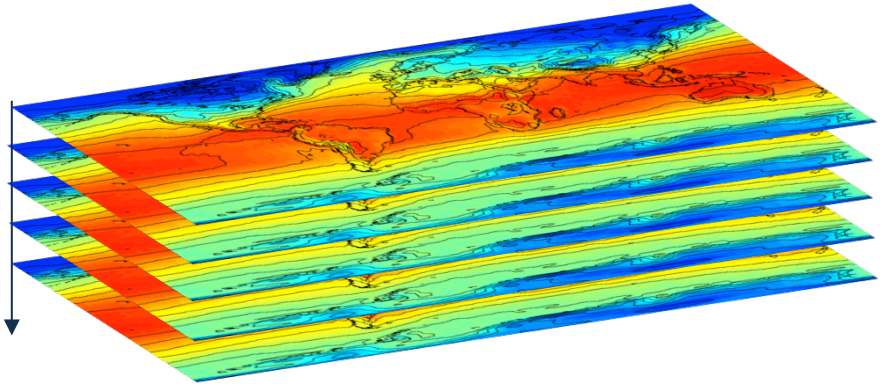
# Exemple of application of SOCA-BBP

## Climatology of $b_{bp}$ for June at a global scale

Surface ocean color climatologies

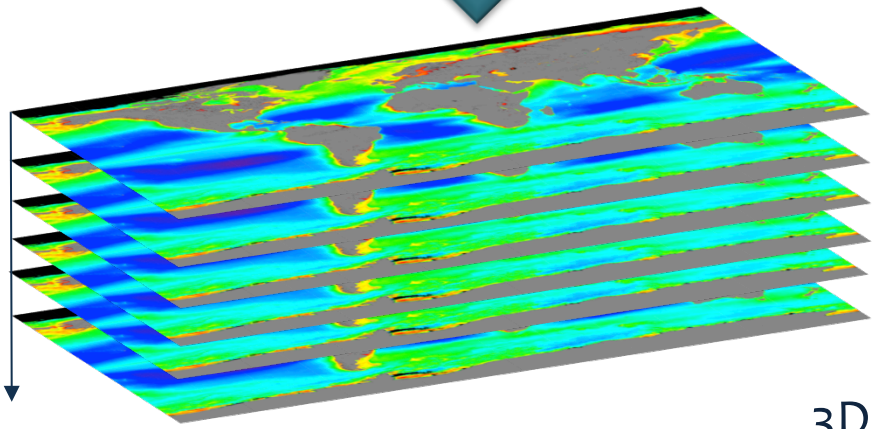


Argo T/S climatology (Roemmich and Gilson ,2009)



+

SOCA-BBP

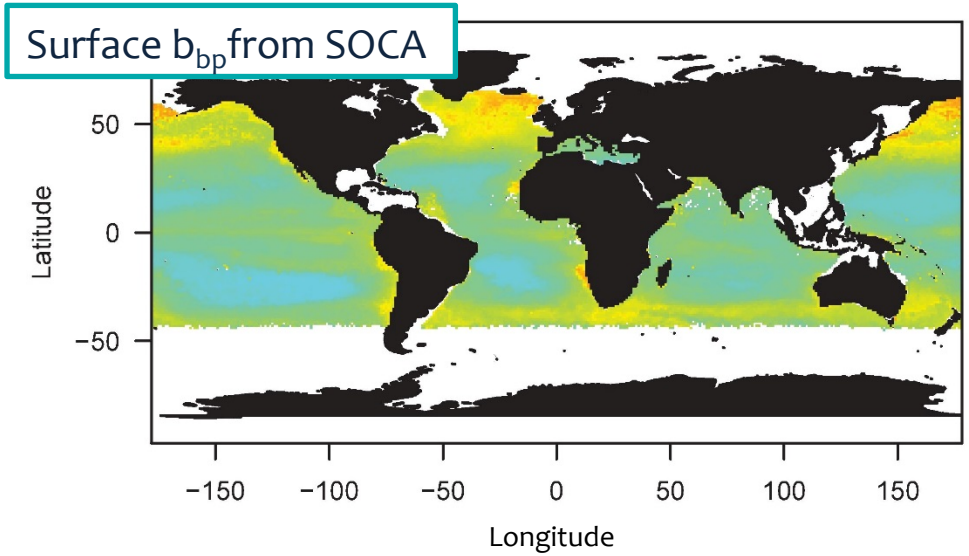


3D global climatology of $b_{bp}$

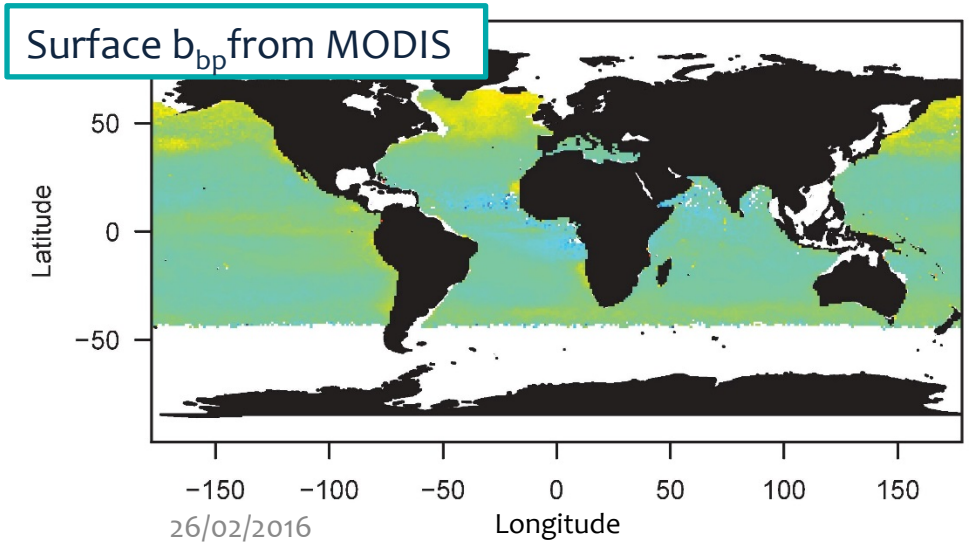
# Exemple of application of SOCA-BBP

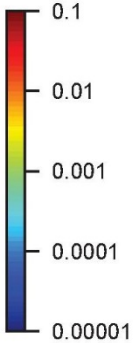
Climatology of $b_{bp}$ for June at a global scale

Surface $b_{bp}$ from SOCA




Surface $b_{bp}$ from MODIS



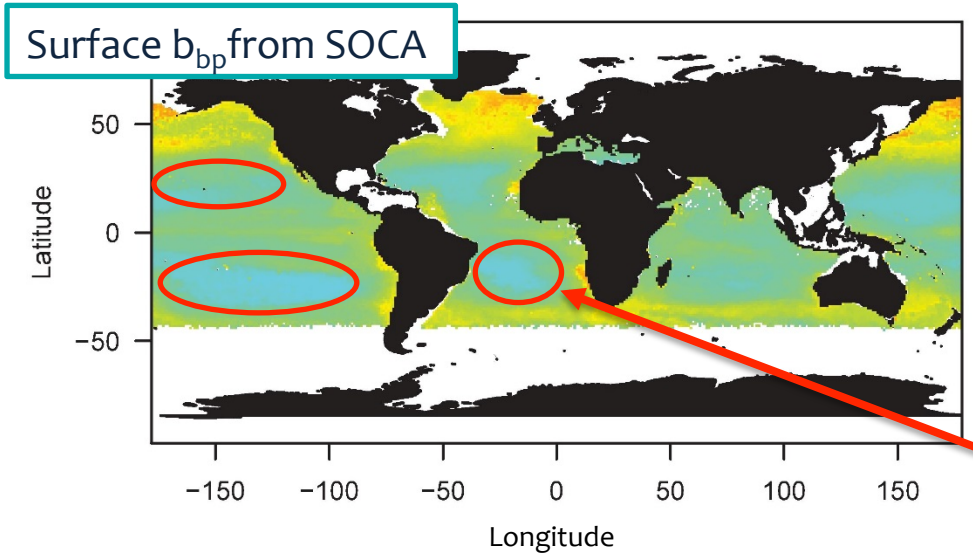



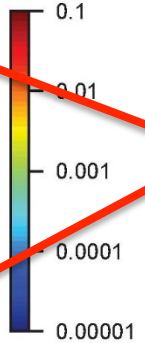

Sauzède *et al.*, In rev.

# Exemple of application of SOCA-BBP

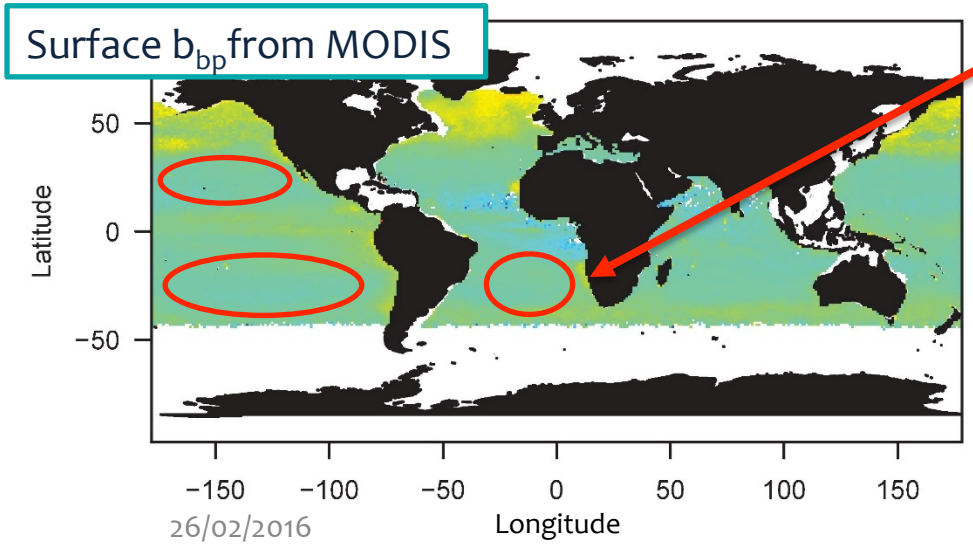
Climatology of $b_{bp}$ for June at a global scale

Surface $b_{bp}$ from SOCA




Surface $b_{bp}$ from MODIS

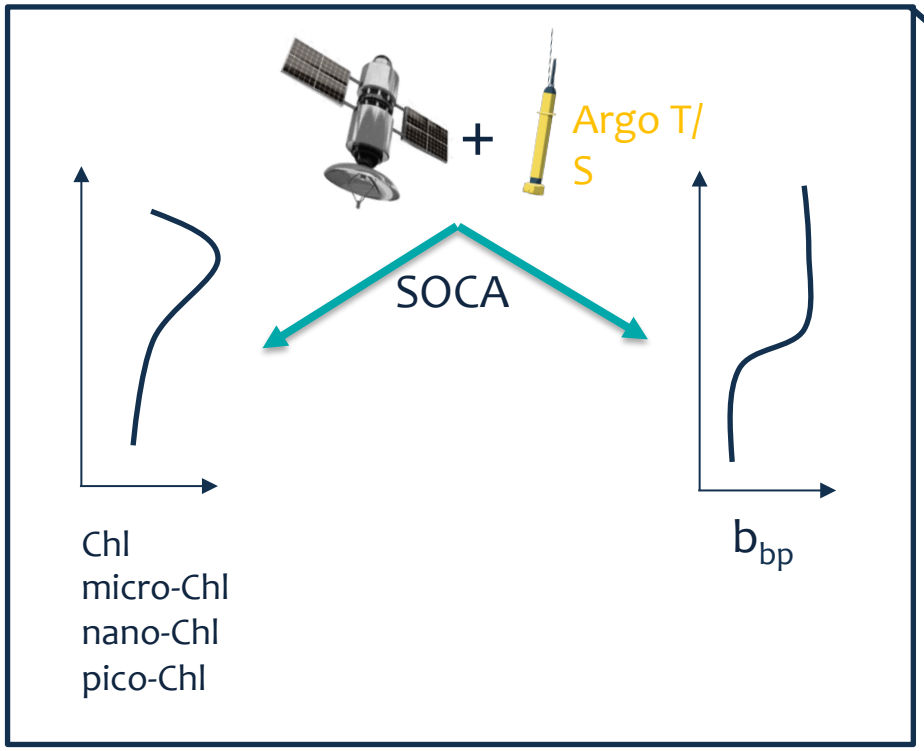


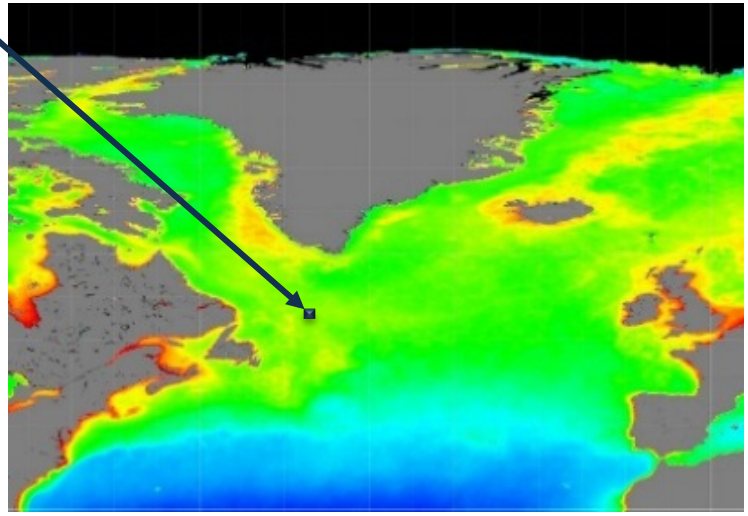

Overestimation of $b_{bp}$ from satellite estimations: e.g. Brown *et al.*, 2008; Lee and Huot, 2014

Sauzède *et al.*, In rev.

# Studying the seasonal dynamics of the vertical distribution of phytoplankton biomass in the North Atlantic

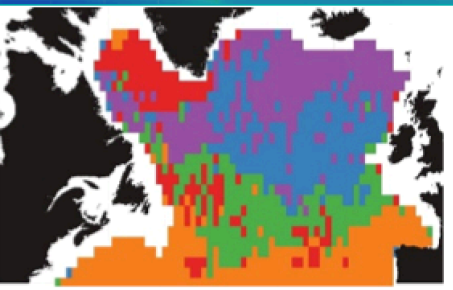
→Specific regional MLP to retrieve the vertical distribution of Chl and the composition of phytoplankton communities (Chl associated to micro-, nano- and pico-phytoplankton)



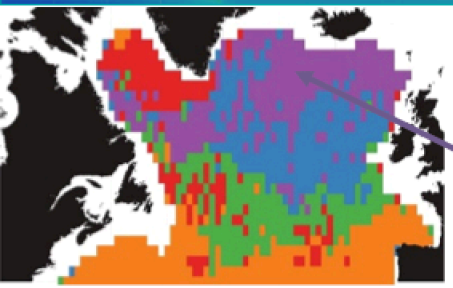
Study based on climatological data

# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic



→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic



→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

**North Islandic & Irminger Sea**

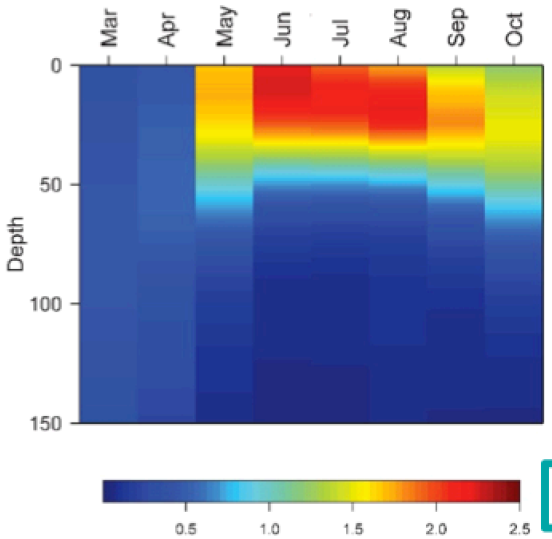
# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic

→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

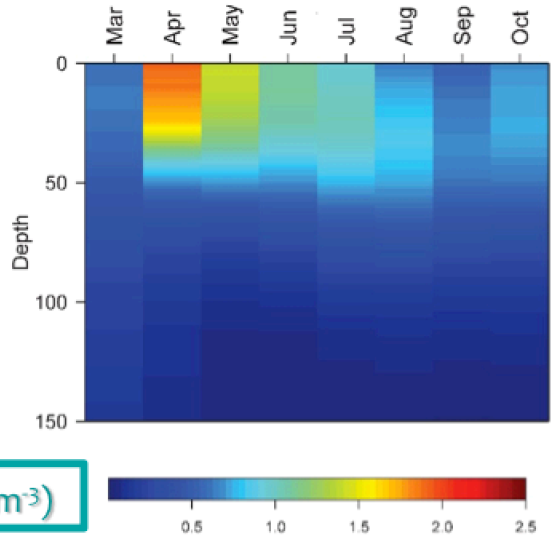
Temperate North Atlantic

→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

**North Islandic & Irminger Sea**



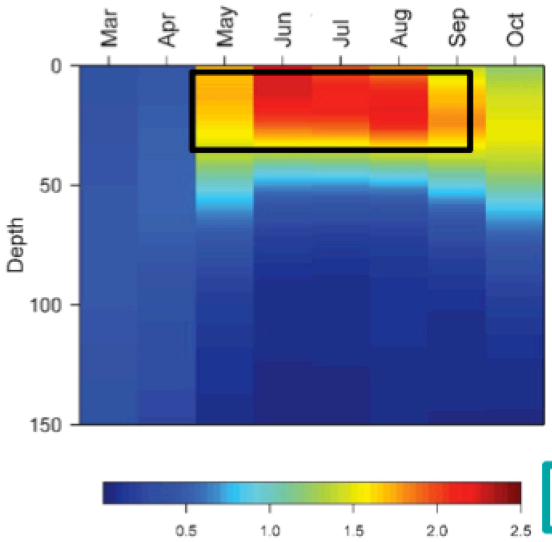

**Temperate North Atlantic**



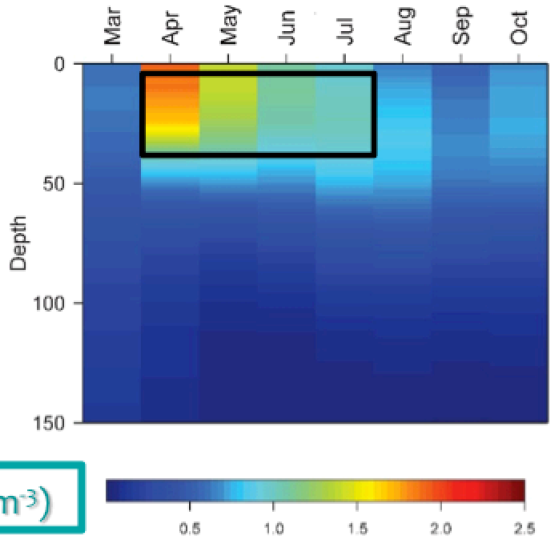

Chl ($mg.m^{-3}$)

→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

**North Islandic & Irminger Sea**



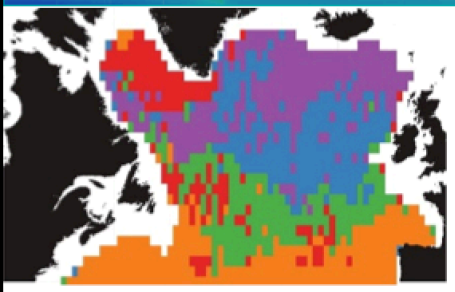

**Temperate North Atlantic**




Chl ($mg.m^{-3}$)

- One month gap in the bloom initiation
- More sustainable Chl in surface layers in the North Islandic and Irminger Seas

# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic



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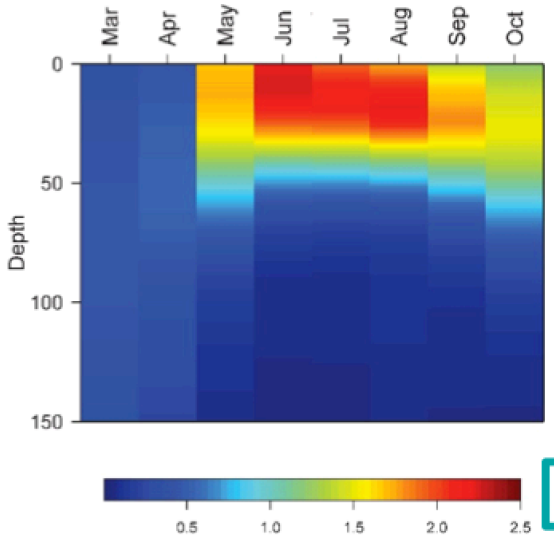
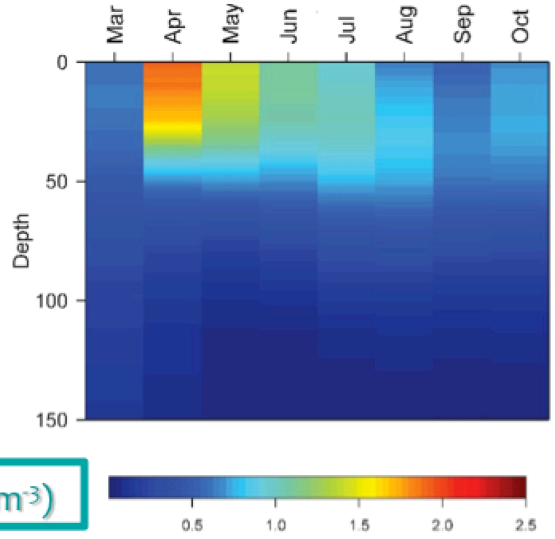
**North Islandic & Irminger Sea**







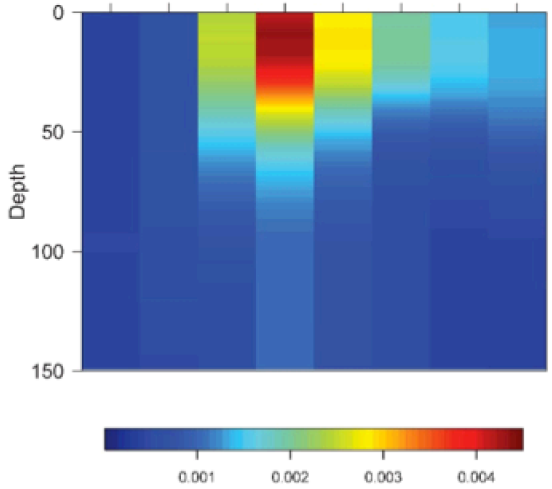
**Temperate North Atlantic**





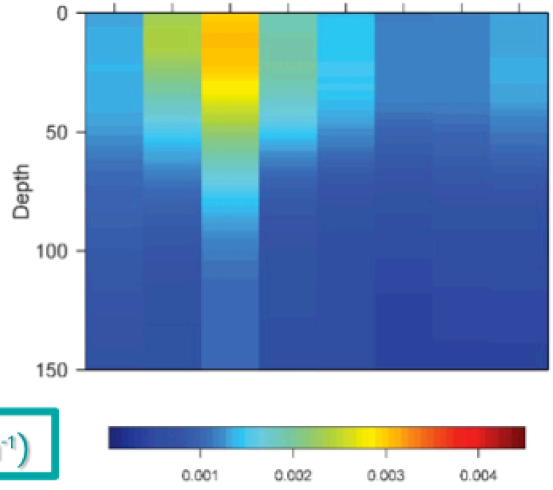


Chl ($mg.m^{-3}$)













$b_{bp}(m^{-1})$

# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic

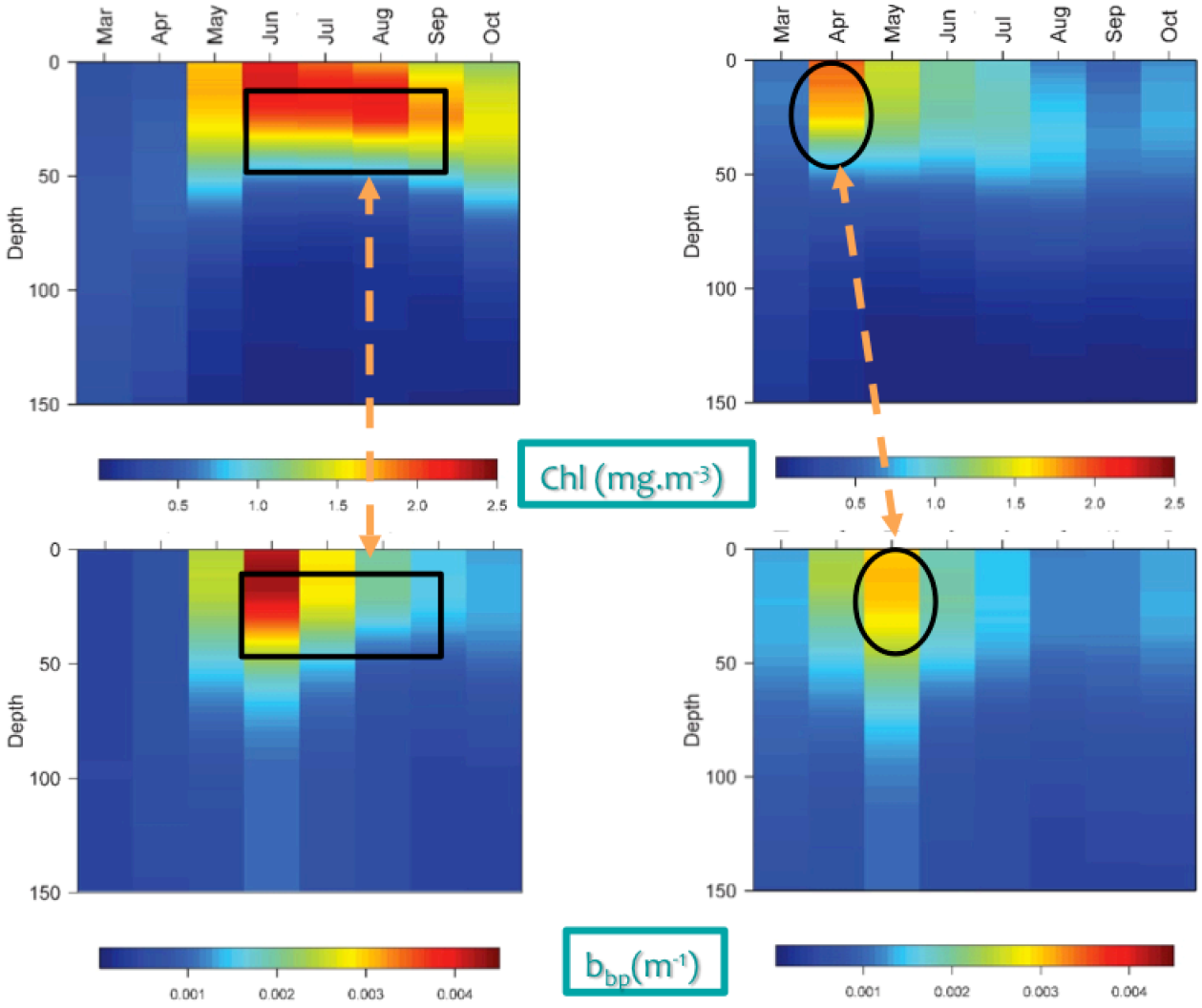
→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

**Change in the ratio Chl/POC:**

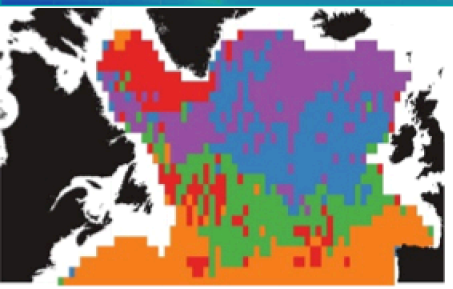
- increase of $b_{bp}$ due to dead phyto-plankton detritus
- decrease of the Chl/POC ratio due to high irradiance (photoacclimation)

North Islandic & Irminger Sea

Temperate North Atlantic



# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic

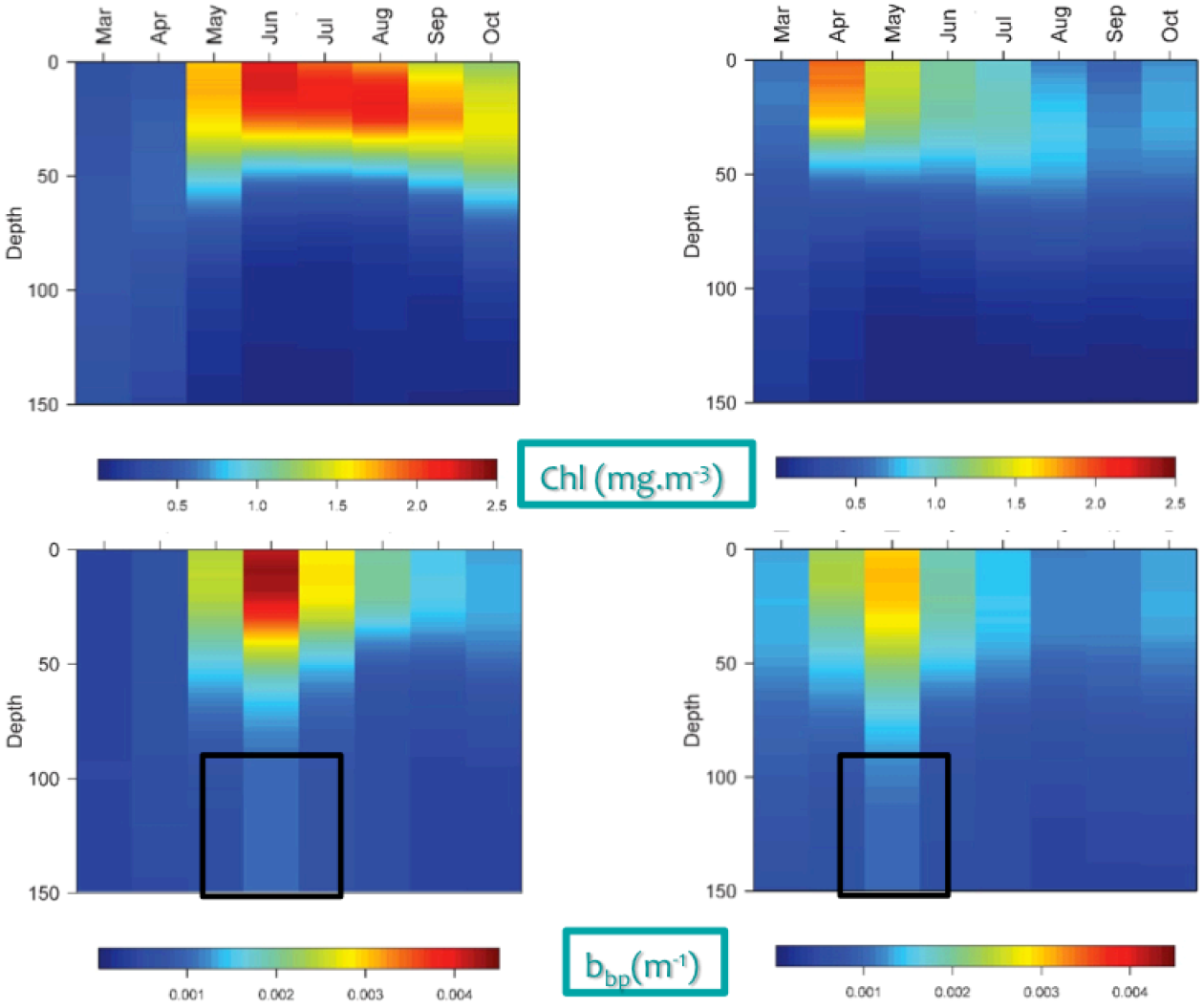


→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

**Significant $b_{bp}$ values at depth:**
Trace of carbon export

North Islandic & Irminger Sea

Temperate North Atlantic

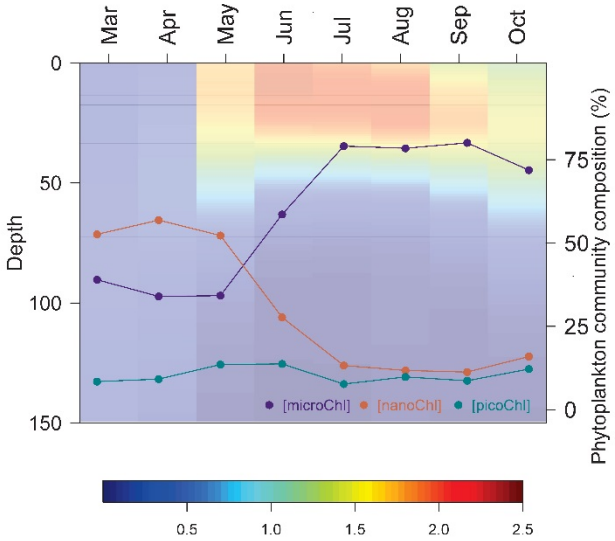


# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic

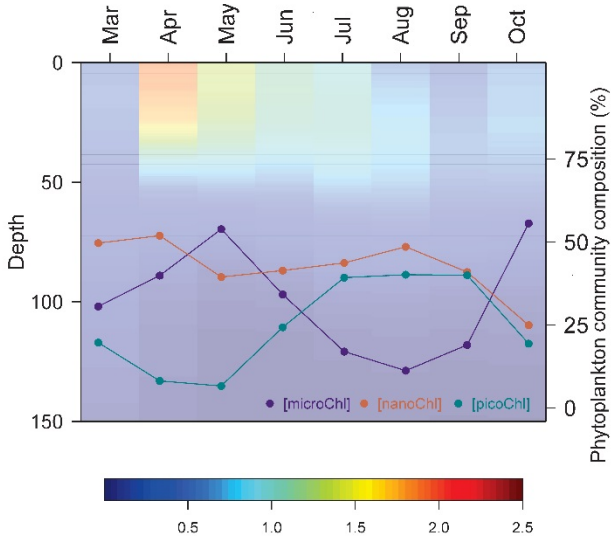
→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

micro-Chl ($mg.m^{-3}$)
nano-Chl ($mg.m^{-3}$)
pico-Chl ($mg.m^{-3}$)

**North Islandic & Irminger Sea**




**Temperate North Atlantic**



# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic

→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

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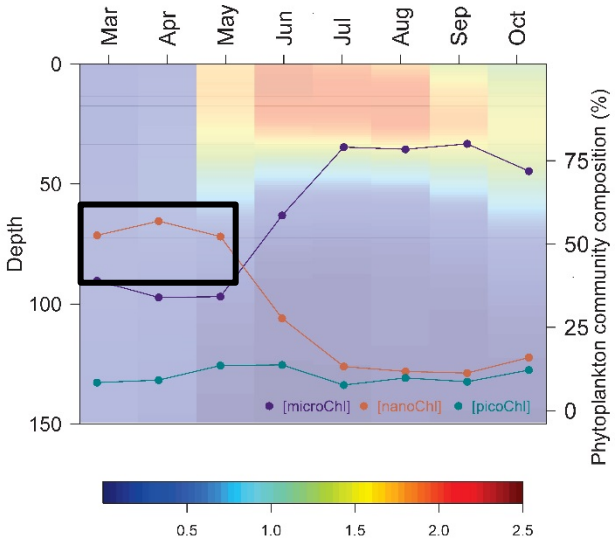
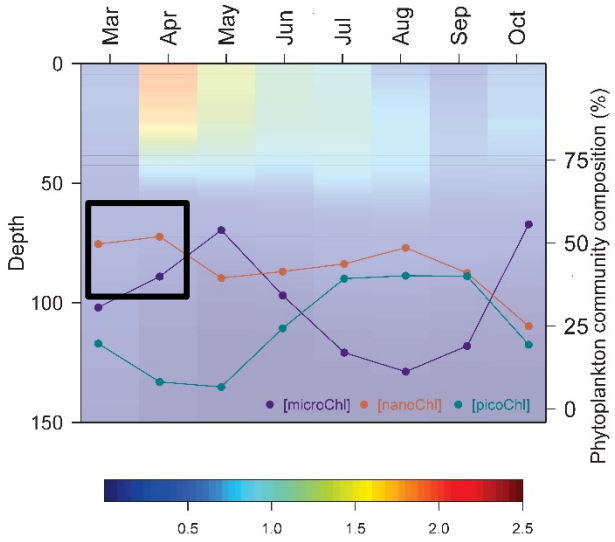
**North Islandic & Irminger Sea**




**Temperate North Atlantic**




**Pre-bloom period:**
Phytoplankton communities dominated by nano-phytoplankton

→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

micro-Chl ($mg.m^{-3}$)
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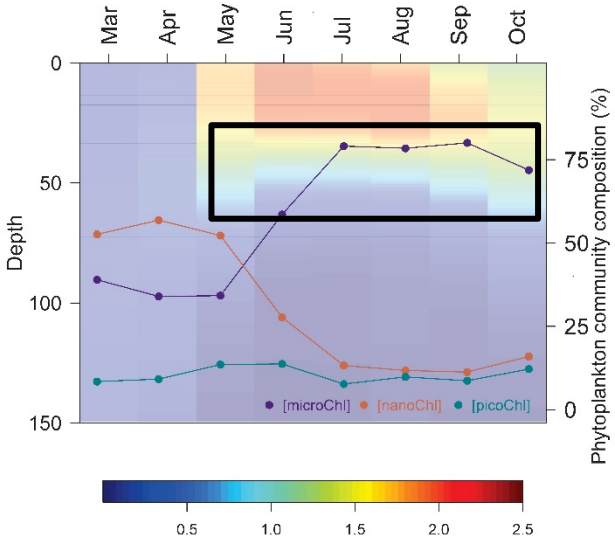
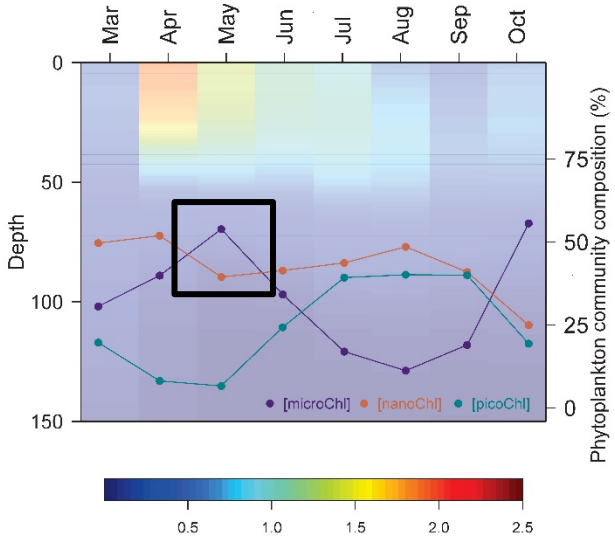
**North Islandic & Irminger Sea**




**Temperate North Atlantic**



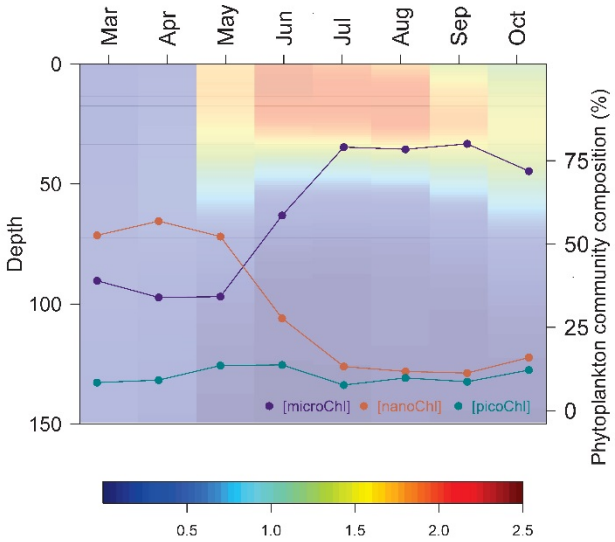

**During bloom period:**
Phytoplankton communities dominated by micro-phytoplankton

# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic

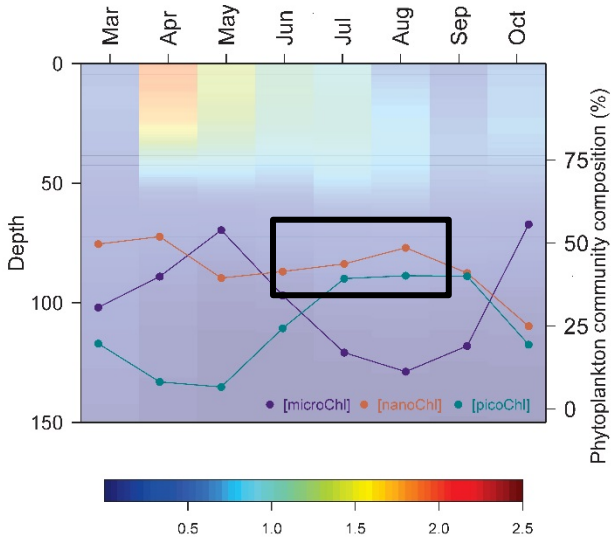
→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

micro-Chl ($mg.m^{-3}$)
nano-Chl ($mg.m^{-3}$)
pico-Chl ($mg.m^{-3}$)

**North Islandic & Irminger Sea**




**Temperate North Atlantic**



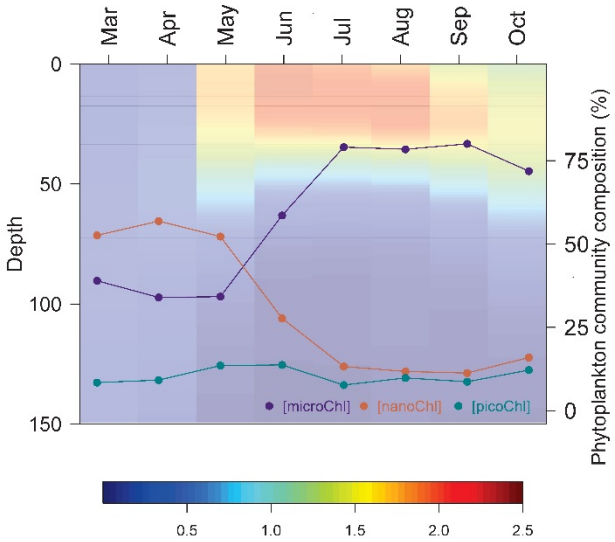

**Temperate North Atlantic post-bloom period:** Phytoplankton communities dominated by nano-phytoplankton

# Vertical distribution of Chl and $b_{bp}$ at a regional scale: a preliminary study in the North Atlantic

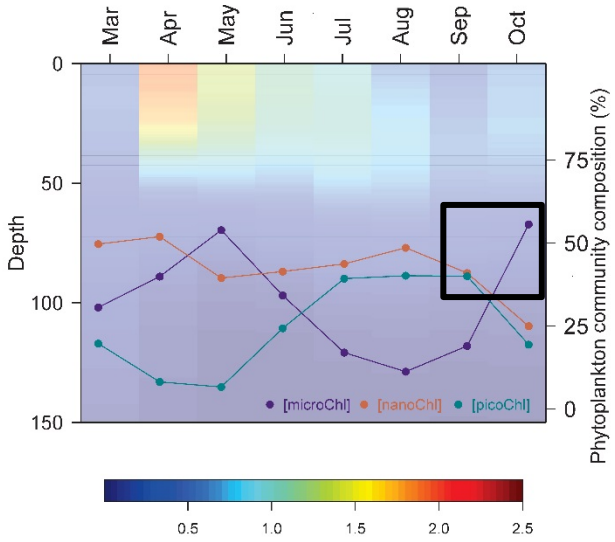
→bio-regionalization using a k-means clusterization (D'Ortenzio and Ribera d'Alcalà, 2009; Lacour *et al.*, 2015) based on the seasonal cycle of normalized surface Chl estimated from satellite data

micro-Chl ($mg.m^{-3}$)
nano-Chl ($mg.m^{-3}$)
pico-Chl ($mg.m^{-3}$)

**North Islandic & Irminger Sea**




**Temperate North Atlantic**



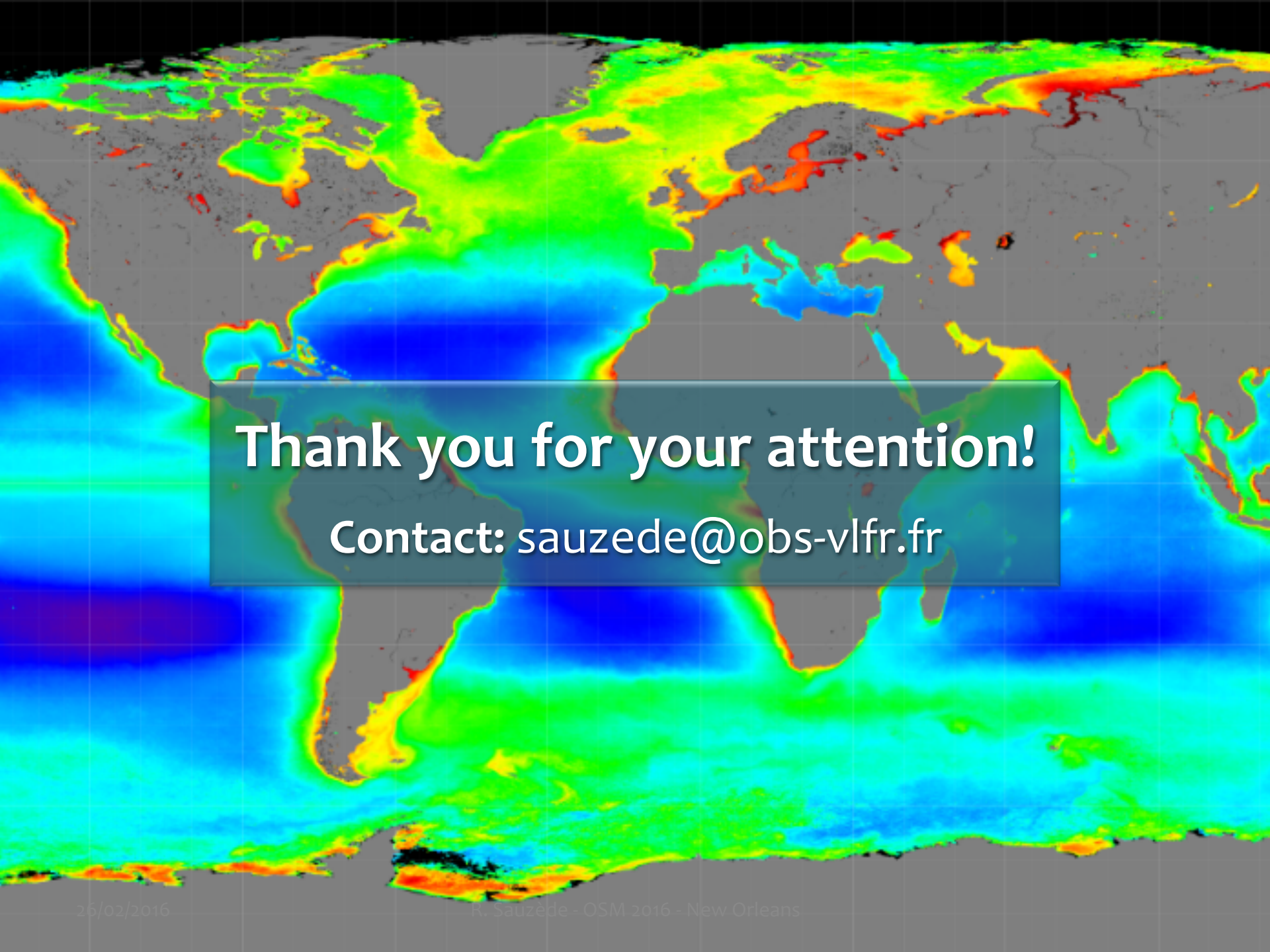

**Temperate North Atlantic:**
Increase of micro-Chl in October due to the fall bloom

# Conclusions and perspectives

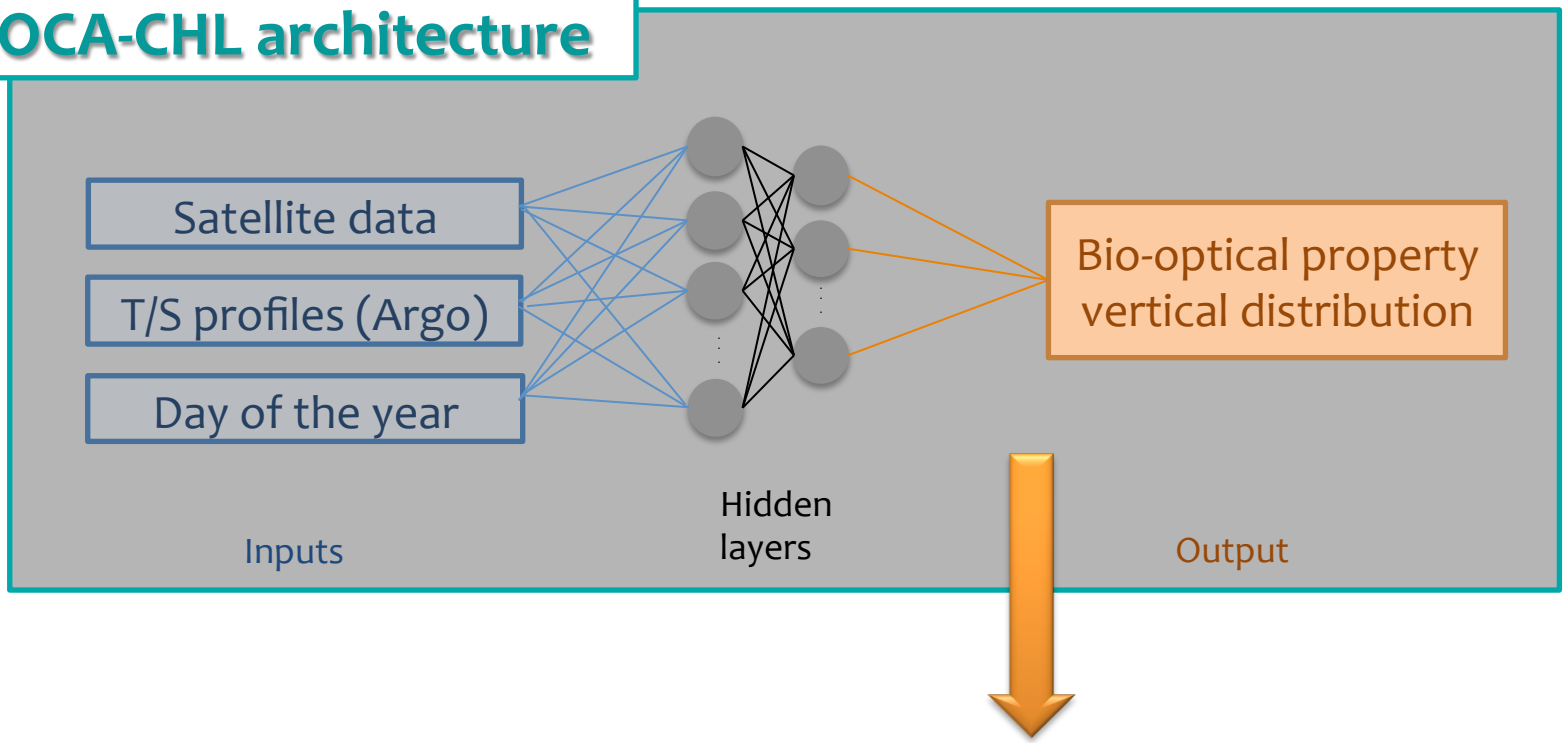
- SOCA: methods that extend surface ocean color bio-optical properties to depth using merged ocean color and Argo data with very promising first results of application at the regional and global scales

- SOCA methods have multiple potential applications such as the creation of 3D climatologies that can be used for the initialization/validation of biogeochemical models or to highlight spatial or temporal anomalies

- Perspective of using both 3D reconstruction of $b_{bp}$ and Chl conjointly: vertical distribution of Chl/POC to better understand light and nutrient control on phytoplankton biomass and physiological status → distribution and variability in primary production and carbon export

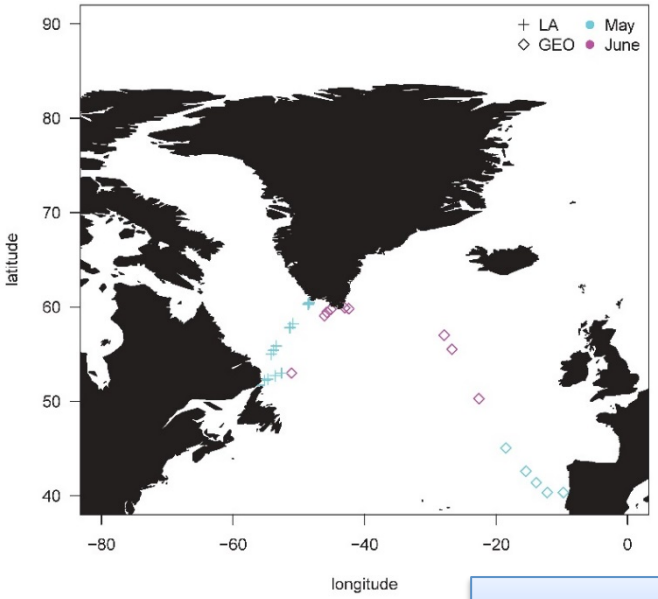

Thank you for your attention!
Contact: sauzede@obs-vlfr.fr

# SOCA-CHL architecture




- **SOCA-CHL:**

→ Chl profile shape (10 normalized Chl values) used to retrieve the Chl and the assemblage of phytoplankton communities using FLAVOR method (Sauzède *et al.*, 2015, JGR)




28 profiles for the validation

| | SOCA-CHL North Atlantic | | | | Uitz *et al.* (2006) model | | | |
|---|---|---|---|---|---|---|---|---|
| | $R^2$ | Slope | MAPD | RMSE | $R^2$ | Slope | MAPD | RMSE |
| Chl | 0.5 | 0.5 | 77 | 2.58 | 0.54 | 0.5 | 68 | 2.93 |
| micro-Chl | 0.38 | 0.59 | 82 | 2.64 | 0.48 | 0.46 | 73 | 3.01 |
| nano-Chl | 0.46 | 0.58 | 63 | 0.14 | 0.27 | 0.4 | 149 | 0.31 |
| pico-Chl | 0.31 | 0.53 | 69 | 0.08 | 0.4 | 0.52 | 124 | 0.1 |